

SINOPSIS KURSUS / COURSE SYNOPSIS

PEJABAT DEKAN FHAS / FHAS DEAN'S OFFICE

FHS4903 Latihan Industri / *Industrial Training*

8 (0+8)

Prasyarat: Telah melengkapkan dan lulus semua kursus termasuk pakej ELEX / *Have completed and passed all courses including the ELEX package*

Melalui kursus ini, pelajar didedahkan kepada persekitaran kerja yang sebenar diindustri/ organisasi. Latihan merangkumi aspek hubung kait teori dan kemahiran praktikal yang telah dipelajari dengan amalan semasa di tempat penempatan latihan industri. Kemahiran komunikasi dan interpersonal ditekankan.

In this course, students are exposed to real working environment in industries/organizations. Training includes application of the theoretical and practical aspects that have been studied with current practices at the place of industrial training placement. Communication and interpersonal skills are emphasized.

JABATAN SAINS PERHUTANAN DAN BIODIVERSITI /
DEPARTMENT OF FORESTRY SCIENCE AND BIODIVERSITY

FHM3100 Sains Perhutanan / Forestry Science

2 (2+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas sains perhutanan dan kepentingannya kepada masyarakat dan negara. Isu perhutanan semasa di peringkat tempatan dan antarabangsa dibincangkan. Kesedaran dan peranan pelbagai pihak dalam memelihara, memulihara dan mengurus hutan secara mampan ditekankan.

This course covers the basic principles of forestry science and its importance to society and the nation. Current forestry issues at the local and international levels are discussed. Awareness and the role of various parties in preserving, conserving and sustainably managing the forests are emphasized.

FHM3102 Botani Hutan / Forest Botany

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi dendrologi dalam perhutanan dan taksonomi tumbuhan. Morfologi untuk pengecaman tumbuhan Gimnosperma dan Angiosperma ditumpukan. Nilai kepelbagaian tumbuhan dan habitatnya dibincangkan.

This course encompasses dendrology in forestry and plant taxonomy. Morphological characters for identification of Gymnosperm and Angiosperm are emphasized. Value of diversities of plants and its habitats are discussed

FHM3103 Dendrologi Dipterokarpa / Dendrology of Dipterocarps

2 (1+1)

Prasyarat: Tiada

Kursus ini merangkumi dendrologi famili pokok Dipterokarpa. Tatanama moden, habitat tumbuhan, kekunci pengecaman, petunjuk botani dan pengumpulan spesimen herbarium ditekankan. Huraian genus dan spesies pokok dipterokarpa serta kepentingannya dalam perhutanan dibincangkan.

This course covers the dendrology of the family Dipterocarpaceae. Modern nomenclature, plant habitats, identification keys, botanical indicators and collection of herbarium specimens are emphasized. Descriptions on genera and species for dipterocarps and its importance in forestry are discussed.

FHM3104 Dendrologi Bukan Dipterokarpa / Dendrology of Non-Dipterocarps

2 (1+1)

Prasyarat: Tiada

Kursus ini merangkumi dendrologi famili pokok bukan Dipterokarpa. Tatanama moden, habitat tumbuhan, kekunci pengecaman, petunjuk botani dan pengumpulan spesimen herbarium ditekankan. Huraian famili dan genus pokok bukan Dipterokarpa serta kepentingannya dalam perhutanan dibincangkan.

This course covers the dendrology of family non- Dipterocarps. Modern nomenclature, plant habitats, identification keys, botanical indicators, and collection of herbarium specimens are emphasized. Description on family and genera for non-Dipterocarps and its importance in forestry is discussed.

FHM3105 Fisiologi Pokok / Tree Physiology

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi peranan, fungsi dan proses fisiologi pokok dalam ekosistem perhutanan. Teknik pengukuran fisiologi pokok yang bersesuaian didedahkan. Hubungan kait antara proses fisiologi pokok dan faktor persekitaran dibincangkan.

This course covers the roles, functions, and processes of tree physiology within forest ecosystem. Appropriate techniques for tree physiological measurement are exposed. The relationship between tree physiological processes and environmental factors is discussed.

FHM3106 Ekologi Hutan / Forest Ecology**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan prinsip ekologi hutan. Teknik ekologi dalam menilai sesebuah jenis hutan ditekankan. Hubung kait ekosistem hutan dan ciri-ciri ekologi hutan dibincangkan

This course covers ecological concepts and forest ecology principles. Ecological techniques in assessing certain forest types are emphasized. Relationships between forest ecosystem and forest ecological characteristics are discussed.

FHM3107 Tanah Hutan / Forest Soil**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi jenis struktur dan sifat tanah. Pengukuran dan pengkelasan mengikut ciri fizikal dan kimia tanah ditekankan. Fungsi dan kepentingan tanah dalam ekosistem hutan serta indeks tapak turut dibincangkan

This course covers the type, structure, and properties of soil. Soil assessment and classification based on physical and chemical properties are emphasized. The functions and importance of soil and site index in the forest ecosystem are discussed.

FHM3108 Entomologi Hutan / Forest Entomology**2 (1+1)**

Prasyarat: Tiada

Kursus ini merangkumi kepentingan serangga dalam ekosistem hutan. Kaedah pengecaman dan pengelasan serangga hutan didedahkan. Kesan perubahan ekosistem hutan kepada kepelbagaian dan fungsi serangga dibincangkan.

This course covers the importance of insects in forest ecosystem. Identification and classification techniques of forest insects are exposed. The effects of forest ecosystem changes on insect biodiversity and functions are discussed.

FHM3109 Patologi Hutan / Forest Pathology**2 (1+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep penyakit pokok, simptom dan tanda penyakit. Penekanan diberi kepada Kursus ini merangkumi konsep penyakit pokok, simptom dan tanda penyakit. Pengenalpastian jenis penyakit berdasarkan patogen pokok hutan ditekankan. Kaedah pengawalan penyakit pokok hutan turut dibincangkan.

This course covers plant disease concept, disease symptoms and signs. Identification of plant disease related to pathogen is emphasized. Control techniques for plant disease management are discussed.

FHM3110 Ekonomi Perhutanan / Forestry Economics**3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teori dalam ekonomi perhutanan. Analisis nilai ekonomi, faedah kos barangan dan perkhidmatan ekosistem hutan ditekankan. Aplikasi teori ekonomi dalam penganggaran nilai perkhidmatan ekosistem hutan dibincangkan.

This course covers concepts and theories in forest ecosystem economic. Economic value analysis, cost- benefit of goods and forest ecosystem services are emphasized. Application of economic theories in estimating the values of forest ecosystem services is discussed.

FHM3111 Kebakaran Hutan / Forest Fire**3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep punca kebakaran hutan dan kesannya kepada alam sekitar. Kaedah pengawalan kebakaran hutan ditekankan. Penilaian terhadap kaedah pencegahan dan pengawalan kebakaran hutan yang sesuai dan berkesan dibincangkan.

This course covers the concept and causes of forest fire and their impact on environment. Methods of forest fire control are emphasized. Assessments of suitable and efficient forest fire prevention and control are discussed.

FHM3112 Kaedah Kuantitatif Dalam Perhutanan / Quantitative Methods in Forestry**4 (3+1)**

Prasyarat: Tiada

Kursus ini merangkumi pemerihalan dan analisis data parametrik dan bukan parametrik dalam perhutanan. Penekanan diberikan kepada pola pembolehubah rawak dan taburan umum parametrik. Teknik persampelan, rekabentuk ujikaji, ralat ujikaji, korelasi dan regresi linear dibincangkan.

This course covers parametric and non-parametric data description and analyses in forestry. The pattern of random variables and some common distributions of parameters are emphasized. Sampling techniques, experimental designs, experimental errors, correlations, and linear regression are discussed.

FHM3207 Survei Hutan / Forest Survey**2 (1+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas survei hutan, teknik ukuran linear dan sudut yang digunakan dalam perhutanan. Kaedah pengukuran kawasan hutan dan penyediaan pelan ditekankan. Teknik pengiraan ketepatan data dan pembetulan yang bersesuaian dibincangkan.

This course covers the basic principles of forest survey, technique of linear and angular measurements which are applied in forestry. Methods of forest area measurement and plan preparation is emphasized. Data accuracy calculation and appropriate corrections are discussed.

FHM3202 Sukatan Hutan / Forest Mensuration**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kepentingan inventori sumber hutan terhadap perancangan pengurusan dan pembangunan hutan. Analisis data inventori sumber hutan yang diperolehi dari aktiviti inventori ditekankan. Jadual stok dan komposisi dirian hutan dibangunkan.

This course covers the importance of forest resource inventory on forest management and development planning. Data analysis of forest resource inventory obtained from inventory activities is emphasized. Table of forest stand stock and composition is established.

FHM3219 Sivikultur Hutan / Forest Silviculture**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep silvikultur dalam sistem pengurusan hutan mampan. Kaedah-kaedah rawatan silvikultur dan pemilihan bahan tanaman didedahkan. Pembangunan rangka kerja silvikultur berdasarkan Sasaran Pembangunan Mampan diterapkan.

This course covers the concept of silviculture in sustainable forest management. Silvicultural treatment and planting material is exposed. The development of a silviculture framework based on the Sustainable Development Goals is applied.)

FHM3204 Kejuruteraan Hutan / Forest Engineering**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas di dalam kejuruteraan hutan yang merangkumi mekanik dan analisis bahan binaan dan infrastruktur hutan, perancangan operasi hutan yang berkekalan dengan mengoptimumkan elemen ekologi, ekonomi dan sosial. Jenis dan penggunaan jentera perhutanan serta kos, penilaian kos alternatif, aspek keselamatan dan aplikasi teknologi terkini dalam kejuruteraan hutan ditekankan.

This course encompasses basic principles in forest engineering, which includes the mechanics and analysis of building materials and forest infrastructure and planning sustainable forest operations by optimizing ecological, economic, and social elements. Types and use of forestry machinery and cost, alternative cost evaluation, safety aspects and applications of the latest technology in forest engineering are emphasized.

FHM3205 Penuaian Hutan / Forest Harvesting**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi aktiviti dan teknik penuaian hutan yang berimpak rendah. Perancangan aktiviti penuaian hutan yang lestari dan berimpak rendah ditekankan. Hubung kait impak penuaian hutan dengan kelestarian pengurusan hutan dibincangkan.

This course covers low impact forest harvesting activities and techniques. Planning on sustainable and low impact forest harvesting activities are emphasized. Relationships between forest harvesting and sustainable forest management are discussed.

FHM3206 Sistem Maklumat Geografi Dalam Perhutanan / Geographic Information System in Forestry**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep asas sistem maklumat geografi (GIS). Kaedah pengendalian alatan dan perisian lembut dalam sistem maklumat geografi didedahkan. Pembuatan keputusan berkaitan perhutanan dengan aplikasi sistem maklumat geografi dibincangkan.

This course covers basic concepts of geographic information system (GIS). Method of instruments handling and software in geographic information system is exposed. Decision making related to forestry with geographic information system application is discussed.

FHM3208 Pembaikbiak Pokok / Tree Improvement**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi analisis kualiti penghasilan dan pengurusan stok tanaman hutan. Pengukuran kualiti pokok dan Teknik pembiakan yang bersesuaian ditekankan. Pelan operasi pembaikbiak pokok yang sesuai dibincangkan.

This course covers quality analysis on production and management of forest planting stocks. Tree quality measurements and suitable breeding techniques are emphasized. Suitable tree improvement operational plan is discussed.

FHM3209 Arborikultur / Arboriculture**3 (2+1)**

Prasyarat: Tiada

Kursus ini mendedahkan teori dan konsep arborikultur terutamanya pengurusan dan penyelenggaraan pokok di kawasan bandar. Konsep asas biologi dan fisiologi pokok, pemilihan tapak dan spesies, kaedah penanaman dan pemangkasan, konsep kesihatan dan penyakit pokok dibincangkan. Aspek profesionalisme dalam pengurusan pokok bandaran melalui amalan terbaik dan pensijilan arboris ditekankan.

This course covers the theory and concept of arboriculture, especially the management and maintenance of trees in urban areas. The basic concept of tree biology and physiology, site and species selection, tree planting and pruning method, tree health and diseases are discussed. Professionalism aspect in urban tree management through best practices and arborist certification is emphasized.

FHM3210 Etnobotani / Ethnobotany**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep kepelbagaian tumbuhan ubatan dan kepentingan kepada manusia. Ciri pokok ubatan bagi tujuan pengecaman dan kegunaan tempatan ditekankan. Hubungkait di antara kepentingan tumbuhan ubatan dan masyarakat setempat dibincangkan.

This course covers the concept of medicinal plants diversity and their importance to humans. Medicinal plant characteristics for identification purposes and local use are emphasized. The link between the importance of medicinal plants and the local community is discussed..

FHM3211 Pemulihan Ekosistem / Ecosystem Restoration**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan prinsip pemulihan ekosistem dan penilaian faktor yang menyebabkan kemerosotan ekosistem hutan. Penilaian dan penggunaan kaedah bagi pemulihan pelbagai ekosistem berhutan yang rosak ditekankan. Keberkesanan aktiviti pemulihan ekosistem dibincangkan.

This course covers the concepts and principles of ecosystem restoration and evaluating factors affecting forest ecosystem degradation. Assessment and application of methods to restore several forest ecosystems is emphasized. Effectiveness of ecosystem restoration activity is discussed.

FHM3212 Kaedah Pensampelan Hidupan Liar / Wildlife Sampling Methods**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknik pensampelan, hidupan liar. Kaedah pensampelan mengikut kumpulan hidupan liar berbeza secara langsung dan tidak langsung ditekankan. Projek lapangan yang melibatkan kaedah pensampelan hidupan liar dibincangkan.

This course covers wildlife sampling concepts and appropriate techniques. Direct and indirect sampling methods of different wildlife groups are emphasized. Field projects involving wildlife sampling methods are discussed.

FHM3213 Perhutanan Tani / Agroforestry**3 (2+1)**

Prasyarat: Tiada

Kursus ini memperkenalkan prinsip dan amalan perhutanan tani di kawasan tropika. Pengajaran akan Kursus ini merangkumi amalan sistem perhutanan tani yang lestari. Penilaian sistem perhutanan tani yang selari dengan kepentingan sosio-ekonomi dan persekitaran ditekankan. Pelan pembangunan dan aktiviti perhutanan tani dibincangkan.

This course covers agroforestry systems based on sustainable practice. The assessment of agroforestry systems in line with socio-economic and environmental interests is emphasized. Development plans and agroforestry activities are discussed.

FHM3214 Teknologi Biji Benih / Seed Technology**2 (1+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknologi biji benih dalam penghasilan dan pengurusan stok tanaman yang mampan. Teknik penghasilan biji benih dan stok tanaman berkualiti ditekankan. Penilaian keperluan penyimpanan dan penghasilan biji benih mengikut kualiti dibincangkan.

This course covers the concept and technology of seeds in the production and management of sustainable plant stocks. Quality seed production techniques and plant stock are emphasized. Assessment of seed storage and production requirements by quality is discussed.

FHM3215 Infrastruktur Hijau Dalam Perhutanan Bandar / Green Infrastructure in Urban Forestry 2 (1+1)

Prasyarat: Tiada

Kursus ini merangkumi fungsi dan komponen infrastruktur hijau dalam konteks perhutanan bandar. Teknik pengukuran komponen dan faedah infrastruktur hijau dalam perhutanan bandar ditekankan. Kesesuaian infrastruktur hijau dalam kawasan perbandaran dibincangkan.

This course covers the functions and components of green infrastructure in the context of urban forestry. Techniques for measuring the components and benefits of green infrastructure in urban forestry are emphasized. The suitability of green infrastructure in urban areas is discussed.

FHM3216 Ekonomi Persekitaran Dan Hutan Bandar / Environmental Economics and Urban Forestry 2 (2+0)

Prasyarat: Tiada

Kursus ini merangkumi isu ekonomi berkaitan hutan bandar. Penilaian faedah persekitaran hutan bandar untuk menangani isu pengurusan hutan bandar ditekankan. Peranan ekonomi persekitaran dalam pengurusan hutan bandar dibincangkan.

This course covers economic issues related to urban forests. Valuation of the environmental benefits on urban forests to address urban forest management issues is emphasized. The role of environmental economics in urban forest management is discussed.

FHM3301 Pengurusan Hutan / Forest Management 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi fungsi dan kepentingan pengurusan hutan secara berkekalan di Malaysia. Amalan dan piawaian pengurusan hutan yang terbaik berkaitan dengan hasil berkekalan dibincangkan. Isu berkaitan dengan pensijilan dan pengurusan hutan secara mampan ditekankan.

This course covers the functions and importance of sustainable forest management in Malaysia. The principles and standard of best forest management practices related to sustainable yield are discussed. Issues related to forest certification and management are emphasized..

FHM3302 Pengurusan Hutan Bandar / Urban Forest Management 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi kaedah amalan baik penanaman, penyelenggaraan dan perancangan pokok di kawasan bandar. Penilaian perancangan pembangunan kawasan hijau di kawasan perbandaran berdasarkan kepada amalan pembangunan mampan ditekankan. Kesesuaian kemudahan dan keperluan sumber manusia dalam aktiviti pengurusan hutan bandar yang mampan dibincangkan.

This course covers good practice methods for tree planting, maintaining and planning in urban areas. Evaluation of green area development planning in urban areas based on sustainable development practices is emphasized. Suitability of green, infrastructure and human resources in sustainable urban forest management activities are discussed.

FHM3303 Pengurusan Hidupan Liar / Wildlife Management 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi implikasi aktiviti perhutanan terhadap hidupan liar. Penilaian faktor bersandar dan tidak bersandar populasi hidupan liar ditekankan. Prinsip pengurusan habitat dan populasi hidupan liar dibincangkan.

The course covers the implications of forestry activities on wildlife. Assessment on wildlife population density dependent and independent factors are emphasized. Principles on wildlife habitat and population management are discussed.

FHM3312 Perhutanan Ladang / Forest Plantation**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep penubuhan ladang hutan khususnya dalam merancang dan membangunkan ladang hutan ke arah Matlamat Pembangunan Mampan (SDGs). Pengaplikasian dan kemahiran teknikal berdasarkan amalan baik tapak samaian dan kaedah penanaman ladang hutan, serta analisis kewangan dipraktikkan. Penilaian isu semasa dan implikasinya kepada pematuhan perundangan, pensijilan dan perjanjian antarabangsa ditekankan.

This course encompasses the concept of forest plantation establishment and focuses on planning and developing forest plantations towards Sustainable Development Goals (SDGs). Application and technical skills based on best practices of good nursery and forest plantation planting techniques and financial analysis are practised. Assessment of current issues and their implications on legal compliance, certification and international agreements are emphasized.

FHM3114 Produk Hutan dan Pengusahaan / Forest Products and Production**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi penerangan jenis dan faedah sumber semula jadi yang diperolehi dari hutan. Respon kepada pemprosesan hasil hutan secara mampan yang menjadi produk komersial ditekankan. Gabungan teknologi, idea dan ciri terbaharu untuk tujuan pengusahaan produk hasil hutan turut dibincangkan.

This course covers the types and benefits of natural resources obtained from the forest. Response on sustainable forest produce processing to be a commercial product is emphasized. The latest combination of technologies, ideas and features for the purpose of processing forest products are discussed.

FHM3315 Pengurusan Kawasan Legeh / Watershed Management**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi amalan pengurusan kawasan tadahan air berdasarkan jenis guna tanah. Analisis dan penilaian data kitaran hidrologi dan hubungan dengan jenis guna tanah ditekankan. Penyelesaian masalah pengurusan kawasan legeh dibincangkan.

The course covers water catchment management practices based on land use. Hydrological data analysis and evaluation and its relation to land use types are emphasized. Solutions for water catchment management problems are discussed.

FHM3316 Hutan dan Perubahan Iklim / Forests and Climate Change**2 (2+0)**

Prasyarat: Tiada

Kursus ini merangkumi peranan hutan dalam mitigasi dan adaptasi perubahan iklim. Peranan pokok dan hutan dalam membina daya tahan terhadap perubahan iklim ditekankan. Respon kepada perubahan iklim, penyelesaian praktikal, cabaran dan isu serta dasar yang diamalkan untuk menangani perubahan iklim dibincangkan.

This course covers the role of forests in climate change mitigation and adaptation. The role of trees and forests in building resilience to climate change are emphasized. Responses to climate change, practical solutions, challenges and issues as well as policies adopted to address climate change are discussed.

FHM3318 Perhutanan Sosial dan Komuniti / Social Forestry and Community**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi peranan hutan sumber hutan sebagai sumber sosio-ekonomi masyarakat. Interaksi baik tentang amalan perhutanan sosial dengan masyarakat ditekankan. Penghargaan terhadap kepentingan dan peranan sumber hutan dalam meningkatkan kesejahteraan dan kehidupan dibincangkan.

This course covers the role of forest resources as a socio- economic resource for society. Good interaction on social forestry practices with the community is emphasized. Appreciation of the importance and role of forest resources in improving well-being and livelihood is discussed.

FHM3306 Undang-Undang Dalam Perhutanan / Laws in Forestry**3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi peruntukan undang-undang dalam sektor perhutanan. Penggunaan undang-undang dalam menangani masalah semasa pengurusan hutan ditekankan. Perbezaan undang-undang di Semenanjung Malaysia, Sabah dan Sarawak dibincangkan.

This course covers provisions of laws in forestry sectors. The application of laws in addressing current forest management issues is emphasized. Differences between laws in Semenanjung Malaysia, Sabah and Sarawak are discussed.

FHM3314 Penyelenggaraan Dan Pengurusan Tapak Semaian / Nursery Management and Maintenance**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi amalan baik kaedah pengurusan dan penyelenggaraan tapak semaian. Isu dan penyelesaian pengurusan dan penyelenggaraan tapak semaian ditekankan. Teknik penggandaan tumbuhan untuk pemuliharaan dan pemasaran dibincangkan.

This course covers the best practices in nursery management and maintenance. Nursery management and maintenance issues and solutions are emphasized. Plant multiplication techniques for conservation and marketing are discussed.

FHM3317 Pengurusan Bersepadu Perosak Pokok Hutan / Integrated Pest Management of Forests Tress**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kaedah pengurusan bersepadu bagi pokok hutan secara sistematik. Penilaian terhadap kesan kerosakan pokok hutan berdasarkan kategori serangga perosak ditekankan. Aplikasi sistem pengurusan perosak bersepadu bagi langkah mengawal serangan serangga perosak dibincangkan.

This course covers systematic methods of integrated management of forest trees. Assessment of the impact on forest trees damages based on insect pests category is emphasized. The application of an integrated pest management system as control measures for pest infestations is discussed.

FHM3311 Pengurusan Kesihatan Pokok / Tree Health Management**2 (1+1)**

Prasyarat: Tiada

Kursus ini merangkumi pengenalanpastian ejen dan penyebab pada penyakit pokok hutan bandar. Penilaian terhadap simptom penyakit pokok hutan bandar berdasarkan kepada ciri-ciri kerosakan ditekankan. Kaedah pengawalan penyakit pokok hutan bandar dibincangkan.

This course covers the identification of agents and causes of urban forest tree diseases. Assessment of urban forest tree disease symptoms based on signs of infestation are emphasized. Methods of urban forest tree disease control are discussed.

FHM4101 Biodiversiti dan Pemuliharaan / Biodiversity and Conservation**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan prinsip biodiversiti dan pemuliharaan bagi suatu ekosistem. Teknik yang bersesuaian untuk mengukur biodiversiti ditekankan. Penilaian biodiversiti bagi sesuatu ekosistem dibincangkan.

This course covers the concepts and principles of biodiversity and conservation for an ecosystem. Appropriate techniques for measuring biodiversity are emphasized. Biodiversity assessment for an ecosystem is discussed.

FHM4102 Ekologi Komuniti Hidupan Liar / Wildlife Community Ecology 3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi konsep struktur dan dinamik komuniti hidupan liar. Pentaksiran terhadap kesan perubahan faktor persekitaran kepada komuniti hidupan liar ditekankan. Kajian kes berkaitan ekologi komuniti hidupan liar dengan aspek konservasi dibincangkan.

This course covers the concepts of wildlife community structure and dynamics. Assessment on the impacts of environmental changes on wildlife communities is emphasised. Case studies related to wildlife community ecology from conservation aspects are discussed.

FHM4103 Ekologi Perilaku Hidupan Liar / Wildlife Behavioural Ecology 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi teori dan konsep asas dalam ekologi perilaku hidupan liar. Ini termasuk mekanisma perilaku hidupan liar yang berkaitan seperti kemandirian, komunikasi, persaingan sumber, mangsa pemangsa dan pergerakan dan keperluan dari segi ruang. Aplikasi ekologi perilaku dalam konservasi hidupan liar ditekankan.

This course covers the theory and fundamental concepts in behavioral ecology of wildlife. It includes mechanisms of wildlife behavior in relation to survival, communication, resource competition, prey and predator, and movement, and space necessity. The applications of behavioral ecology in wildlife conservation are emphasized.

FHM4201 Penderiaan Jauh Dalam Perhutanan / Remote Sensing in Forestry 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi integrasi keperluan teknologi foto udara dan penderiaan jauh dalam isu perhutanan. Penaksiran imej foto udara berdasarkan situasi penyelesaian masalah ditekankan. Aplikasi teknologi penderiaan jauh dalam pengurusan hutan turut dibincangkan.

This course covers the integration of aerial photography and remote sensing technology requirements in forestry issues. Evaluation of aerial photo images based on problem-solving situations is emphasized. The application of remote sensing technology in forest management is also discussed.

FHM4202 Penilaian Ekonomi Sumber Hutan Dan Biodiversiti / Economic Valuation of Forest Resource and Biodiversity 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi teori dan konsep penilaian ekonomi mengikut jenis sumber hutan dan kepelbagaian biologi. Teknik penilaian ekonomi sumber hutan dan biodiversiti yang sesuai ditekankan. Nilai ekonomi sumber hutan bagi ekosistem yang berlainan dibincangkan.

This course covers the theory and concept of economic valuation according to the type of forest resources and biological diversity. Appropriate economic valuation techniques of forest resources and biodiversity are emphasized. Economic values of forest resources for different ecosystems are discussed.

FHM4203 Perhutanan Antarabangsa / International Forestry 2 (2+0)

Prasyarat: Tiada

Kursus ini merangkumi status, klasifikasi hutan dunia dan isu semasa global berkaitan dengan sektor perhutanan. Perbincangan berkisar kepada organisasi perhutanan global dan nasional dan perjanjian antarabangsa serta kaitannya dengan pengurusan hutan.

This course covers status, classification of forest and current global issues related to forestry sectors. Discussion on global and national forestry organizations, and international treaty related to forest management.

FHM4204 Aplikasi Bioteknologi Dalam Perhutanan / *Biotechnology Application in Forestry* 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan kaedah bioteknologi dalam pengurusan hutan mampan. Teknik molekular dalam identifikasi spesies flora dan fauna ditekankan. Kaedah bioteknologi yang boleh digunapakai dalam pengurusan hutan mampan dibincangkan.

This course covers the principles and methods of biotechnology in sustainable forest management. Molecular techniques in the identification of flora and fauna species are emphasized. Biotechnological methods that can be used in sustainable forest management are discussed.

FHM4205	Teknik Hidrologi Hutan / <i>Forest Hidrology Technique</i>	3 (2+1)
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Prasyarat: Tiada

Kursus ini merangkumi aktiviti pengukuran proses hidrologi menggunakan peralatan yang bersesuaian. Hubungan antara data hidrologi dengan proses hidrologi hutan ditekankan. Teknik kajian hidrologi dan analisis data dibincangkan.

This course covers the measurement activity of hydrological processes using appropriate equipment. The relationship between hydrological data and forest hydrological processes is emphasized. Hydrological study techniques and data analysis are discussed.

FHM4206	Pengukuran Dan Pemantauan Karbon / <i>Carbon Measurement and Monitoring</i>	3 (2+1)
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Prasyarat: Tiada

Kursus ini merangkumi penilaian kadar perubahan karbon di pelbagai jenis hutan. Teknik pengukuran anggaran karbon ditekankan. Isu karbon hutan dengan perubahan iklim dibincangkan.

This course covers the assessment of carbon change rate in various forest types. Measurement techniques for carbon estimation are emphasized. Forest carbon issues and climate change are discussed.

FHM4207	Ekologi Perbandaran / <i>Urban Ecology</i>	3 (2+1)
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Prasyarat: Tiada

Kursus ini merangkumi komponen ekosistem perbandaran mampan. Permasalahan dan penyelesaian berkaitan ekologi perbandaran ditekankan. Strategi penyelesaian permasalahan ekologi perbandaran dibincangkan.

This course covers the components of a sustainable urban ecosystem. Problems and solutions related to urban ecology are emphasized. Strategies for solving urban ecological problems are discussed.

FHM4301	Pengurusan Hutan Lanjutan / <i>Advance Forest Management</i>	2 (1+1)
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Prasyarat: Tiada

Kursus ini merangkumi prinsip perancangan yang berasaskan matlamat dan objektif pengurusan hutan mampan. Teknik perancangan yang sesuai dalam pelaksanaan amalan pengurusan hutan mampan ditekankan. Penilaian terhadap pelbagai fungsi objektif termasuk masalah bagi pembangunan pelan pengurusan hutan dibincangkan.

This course covers planning principles based on the goals and objectives of sustainable forest management. Appropriate planning techniques in the implementation of sustainable forest management practices are emphasized. Assessment of the multiple objective functions including problems for the development of forest management plans are discussed.

FHM4302	Konflik Manusia Dan Hidupan Liar / <i>Human and Wildlife Conflicts</i>	3 (3+0)
Prasyarat:	Tiada	
Kursus ini merangkumi konsep dalam memahami punca dan penyelesaian konflik manusia dan hidupan liar. Kaedah pengukuran tahap konflik pengguna langsung dan tidak langsung ke atas hidupan liar ditekankan. Nilai sosio-ekonomi dan alam sekitar dalam konflik manusia dan hidupan liar dibincangkan. <i>This course covers concepts in understanding the causes and solutions of human and wildlife conflict. Methods of measuring the level of direct and indirect user conflict over wildlife are emphasized. Socio-economic and environmental values in human-wildlife conflict are discussed.</i>		
FHM4303	Pengurusan Risiko Dalam Hutan Bandar / <i>Risk Management in Urban Forestry</i>	3 (2+1)
Prasyarat:	Tiada	
Kursus ini merangkumi konsep dan kaedah pengurusan risiko dalam pengurusan hutan bandar. Langkah yang bersesuaian bagi mengurangkan risiko di hutan bandar ditekankan. Pelan pengurusan risiko yang bersesuaian dalam pengurusan hutan bandar dibincangkan. <i>This course covers the concepts and methods of risk management in urban forestry. Appropriate measures to reduce risk in urban forests are emphasized. Suitable risk management plans in urban forest management are discussed.</i>		
FHM4904	Kem Perhutanan / <i>Forestry Camp</i>	2 (0+2)
Prasyarat	Tiada	
Kursus ini mendedahkan kepada pelajar berkenaan kemahiran perhutanan dan kelangsungan hidup dalam hutan sebenar. Kemampuan menggunakan sumber yang terhad di persekitaran hutan dengan betul dan selamat ditekankan. Kemahiran pengurusan diri dan kumpulan diterapkan. <i>This course exposes students to forestry skills and survival in real forests. The ability to use limited resources in a forest environment correctly and safely is emphasized. Self- and group management skills are applied</i>		
FHM4959	Disertasi Bachelo / <i>Bachelor Dissertation</i>	6 (0+6)
Prasyarat:	Tiada	
Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai diberi penekanan. <i>This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized</i>		

JABATAN INDUSTRI KAYU DAN GENTIAN /
DEPARTMENT OF WOOD AND FIBER INDUSTRIES

FHH3101 Sains Kayu / Wood Science

3 (2+1)

Prasyarat: Tiada

Kursus ini memperkenalkan kepada pelajar berkenaan kepentingan kayu dalam kehidupan manusia. Pengajaran akan menumpu kepada sifat anatomi, kimia, fizikal, dan mekanikal kayu serta perkaitannya dalam pemprosesan dan penghasilan produk berasas kayu. Pelajar akan didedahkan kepada teknik pengecaman pelbagai spesies kayu komersil.

This course introduces students on the importance of wood to human life. The teaching will focus on the anatomical, chemical, physical, and mechanical properties of wood in relation to the processing and production of wood-based products. The student is exposed to the identification technique of various commercial wood species

FHH3102 Anatomi Kayu / Wood Anatomy

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi kepelbagaian struktur dan ciri-ciri anatomi kayu serta mekanisma pembentukan dan perkembangannya. Prosedur pengecaman spesies kayu secara kaedah makroskopik dan mikroskopik berpandukan kepada sifat anatomi sesuatu sampel kayu didedahkan. Kepentingan struktur dan sifat anatomi untuk penggunaan kayu dibincangkan.

This course covers the various wood anatomical structures and traits as well as their formation and development mechanism. Procedures for the identification of wood species through macroscopic and microscopic methods based on the anatomical properties of a wood sample are exposed. The importance of wood structure and anatomical properties for wood utilization is discussed.

FHH3103 Fizik Kayu / Wood Physics

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi sifat fizik kayu sebagai bahan mentah industri. Ujian dan analisis data berkaitan sifat fizik kayu dipraktikkan. Sifat fizik kayu berpandukan kepada penggunaan standard dibincangkan.

This course covers the physical properties of wood as an industrial raw material. Tests and data analysis related to the physical properties of wood are practiced. The physical properties of wood based on standard usage are discussed.

FHH3104 Kimia Kayu / Wood Chemistry

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi proses pembentukan dan taburan komponen kimia dalam kayu dan gentian. Kandungan bahan kimia dalam kayu dan gentian ditentukan. Tindak balas komponen kimia kayu terhadap asid, alkali, haba, dan sinaran UV dibezakan.

This course covers the process of formation and distribution of chemical components in wood and fiber. The content of chemicals in wood and fiber is determined. The reaction of the chemical components of wood to acid, alkali, heat, and UV radiation is distinguished.

FHH3105 Mekanik Kayu / Wood Mechanic**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi sifat mekanik kayu pejal dan komposit kayu. Konsep dan ujian mekanik kayu, faktor yang mempengaruhi sifat mekanik, dan kaitannya dengan penggunaan ditekankan. Kriteria rekabentuk elastik untuk struktur kayu, papan komposit dan kombinasi kedua-duanya didedahkan.

This course encompasses the mechanical properties of solid wood and composite woods. Concept and testing in wood mechanics, factors that influence the mechanical properties and its relation to utilization are emphasized. Elastic design criteria for wood and wood composite structures and the combination of both are exposed.

FHH3106 Pengenalan Kepada Operasi Hutan / Introduction to Forest Operation**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kaedah dan teknik di dalam sistem operasi hutan. Disamping itu, kaedah survei dan sukatan juga diperkenalkan. Pembinaan pelan operasi hutan, termasuk jalan hutan, pengurusan dan penyelenggaraan jentera didedahkan.

This course encompasses the theory and technique of forest operation systems. Besides, survey and mensuration theory and techniques are introduced. Development of operational plan including forest road, management and maintenance of machinery are exposed.

FHH3201 Hasil Hutan Bukan Kayu / Non-Wood Forest Products**3 (2+1)**

Prasyarat: Tiada

Kursus ini memperkenalkan kepelbagaian hasil hutan bukan kayu. Penentuan spesies dan pengujian sifat hasil hutan bukan kayu dipraktikkan. Isu semasa berkaitan perolehan, pemprosesan, dan pembuatan produk akhir hasil hutan bukan kayu di Malaysia dibincangkan.

This course introduces the diversity of non-wood forest products and their uses. Species identification and testing of the properties of non-wood forest products are practiced. Current issues related to the procuring, processing, and manufacturing of final products of non-wood forest products in Malaysia are discussed.

FHH3202 Pengilangan Dan Pengeringan Kayu / Sawmilling and Wood Drying**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi teori, konsep dan amalan dalam pengilangan dan pengeringan kayu. Peralatan, langkah-langkah pemprosesan, kaedah dan faktor yang mempengaruhi pulangan dalam pengilangan kayu didedahkan. Prinsip dan kaedah pengeringan kayu serta punca dan kawalan kecacatan kayu akibat pengeringan dibincangkan.

This course encompasses theory, concept and practices of sawmilling and wood drying. Machineries, processing steps, methods and factors that affect sawmilling recovery are exposed. Principles and methods of wood drying as well as causes and preventions of drying defects are discussed.

FHH3203 Kemerosotan Kayu dan Rawatan / Wood Deterioration and Treatment**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi aspek kemerosotan kayu dan hasil hutan lain. Punca kemerosotan kayu disebabkan oleh agen biotik dan abiotik dibincangkan. Kaedah melindungi dan mengawet kayu didedahkan.

This course encompasses the aspects of deterioration of wood and other forest products. Biotic and abiotic agents that affected deterioration are discussed. Methods of wood protection and preservation are exposed.

FHH3204 Perekat dan Teknologi Perekatan Kayu / Wood Adhesive and Adhesion Technology 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi pelbagai jenis perekat kayu serta sifatnya. Teori asas dan sifat fizikal perekat serta mekanisma perekatan ditekankan. Aplikasi dan ujian rekatan pelbagai jenis perekat dalam penghasilan komposit kayu dijalankan.

This course encompasses various types of wood adhesives and their properties. Fundamental theory and physical properties of the adhesives and its adhesion mechanism are emphasized. Application and adhesion tests of wood composites are conducted.

FHH3205 Biokomposit / Biocomposites 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi jenis dan proses pembuatan untuk menghasilkan pelbagai panel biokomposit berdasarkan kesesuaian produk akhir. Pelbagai teknik untuk menghasilkan panel biokomposit didedahkan. Pengujian dan penilaian sifat fizikal dan mekanikal panel biokomposit dilaporkan.

This course covers the types and manufacturing processes to produce various biocomposite panels based on the suitability of the final product. Various techniques for producing biocomposite panels are exposed. Testing and evaluation of the physical and mechanical properties of biocomposite panels is reported.

FHH3206 Pemesinan Kayu / Wood Machining 3 (2+1)

Prasyarat: Lulus / Pass FHH3202

Kursus ini merangkumi teori, konsep dan kaedah pemesinan kayu. Hubungkan pemboleh ubah pemesinan dan kualiti pemesinan di dalam proses pemesinan kayu, seperti pengetaman dan pengupasan. Kerja pemesinan kayu dengan membuat perabot ringkas juga dilaksanakan.

The course encompasses theory, concept and methods of wood machining. Relationship of machining variables with performance and quality of machining, especially planing and veneer cutting will be discussed. Wood machining activities by making simple furniture prototypes will be conducted.

FHH3207 Teknologi Pulpa dan Kertas / Pulp and Paper Technology 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi kepelbagaian jenis sumber gentian dan pencirian bahan mentah bagi industri pulpa dan kertas. Teknik mempulpa, proses pelunturan dan penyediaan stok pulpa serta proses pembuatan kertas didedahkan. Perolehan bahan dan pertimbangan alam sekitar dalam industri pulpa dan kertas dibincangkan.

This course encompasses various types of fiber sources and raw material characterization for the pulp and paper industry. Techniques of pulping, bleaching processes, pulp stock preparation and papermaking are exposed. Procurement of materials and environmental consideration in the pulp and paper industry are discussed.

FHH3208 Proses Pembuatan Perabot / Furniture Manufacturing Process 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi latar belakang industri perabot di Malaysia dan seluruh dunia. Prinsip dan kaedah rekabentuk perabot, proses utama pembuatan perabot, isu alam sekitar dan kesihatan yang terlibat ditumpukan. Teknik penilaian kualiti dan produktiviti pembuatan perabot melalui kajian kes dilaksanakan.

This course encompasses the background of Malaysian and global furniture industries. Principles and techniques of furniture design, furniture manufacturing processes, and the related environmental and health issues are emphasized. Assessment techniques on quality and productivity through case studies are conducted.

FHH3209 Aplikasi Komputer Dalam Reka Bentuk Produk Kayu / *Computer Application in Wood Product Design* **3 (1+2)**

Prasyarat: Tiada

Kursus ini memperkenalkan kepada aplikasi komputer dalam reka bentuk produk kayu. Penghasilan modul produk berpandukan kemahiran mereka bentuk berbantuan komputer menggunakan aplikasi teknologi percetakan 3D dipraktikkan. Idea produk berasaskan kayu berpandukan kepada konsep carta alir proses reka bentuk ditekankan.

This course introduces computer applications in wood product design. Production of product modules based on computer-aided design skills using 3D printing technology applications is put into practice. Wood-based product ideas based on the design process flow chart concept are emphasized.

FHH3301 Perolehan Bekalan Hasil Hutan / *Forest Products Procurement* **3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan kaedah perolehan bekalan hasil hutan untuk proses hiliran. Kaedah pengukuran isipadu kayu balak, pemeringkatan kayu gergaji didedahkan. Sistem perolehan bekalan hasil hutan secara sah dan mampan ditekankan.

This course encompasses the concepts and methods of forest products procurement for downstream processing. Log volume measurement and sawn timber grading are exposed. Legal and sustainable procurement system of forest products are emphasized.

FHH3303 Ekonomi Industri Perakayuan / *Wood Industry Economics* **3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip ekonomi mikro seperti pengeluaran, penawaran dan permintaan yang melibatkan industri perakayuan. Teori kos, diskriminasi harga, struktur dan tingkah laku pasaran serta aplikasinya dalam industri perakayuan ditekankan. Isu-isu semasa dalam ekonomi industri perakayuan turut dibincangkan.

This course encompasses the microeconomics principles such as production, demand and supply related to the wood industry. Cost theory, price discrimination, market structure and behavior as well as its application in wood industry are emphasized. Current issues on wood industry economics are also discussed.

FHH3304 Analisis Operasi Industri Hasil Hutan / *Operation Analysis in Forest Products Industries* **3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknik berkaitan perancangan, reka bentuk, dan pengawalan operasi penghasilan produk hasil hutan. Analisis proses pembuatan produk hasil hutan untuk mengoptimumkan kecekapan pengeluaran ditumpukan. Penyelesaian masalah berdasarkan maklumat yang diperolehi dari lapangan dijalankan.

This course encompasses the concept and techniques related to planning, design, and control in forest products manufacturing operations. Analysis of forest products manufacturing process to optimize production efficiency are emphasized. Problem solution based on field information are conducted.

FHH3305 Analisis Kewangan Industri Hasil Hutan / *Financial Analysis of Forest Products Industries* **3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknik pengurusan kewangan. Pengajaran menumpu kepada pemahaman serta penggunaan alatan analitikal yang bersesuaian dalam membuat analisis dan keputusan kewangan dalam industri hasil hutan. Pelajar juga diberi peluang menganalisis penyata kewangan sebenar firma industri hasil hutan.

This course covers the basic concepts and techniques of financial management. Teaching focuses on understanding and use of analytical tools in analysing and making financial decision in the forest products industries. Students are also given the opportunity to analyze actual financial statements of forest products industry firm.

FHH3308 Pemasaran dan Perdagangan Produk Berasaskan Kayu / Marketing and Trading of Wood-Based Product 3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan konsep dalam pemasaran dan perdagangan produk berasas kayu. Analisis pasaran produk berasas kayu di peringkat tempatan dan antarabangsa dibincangkan. Pembangunan pelan pemasaran produk berasas kayu diperkenalkan.

This course covers principles and concepts in the marketing and trading of wood-based products. Market analysis of wood-based products at the local and international levels is discussed. The development of a marketing plan for wood-based products is introduced.

FHH3309 Pengurusan dan Polisi Perakayuan / Forest Industry Policy and Management 3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip serta aktiviti dalam mengurus sesebuah organisasi. Pengurusan sumber manusia, pengeluaran, pemasaran dan kewangan serta perkaitannya dengan pengurusan sesebuah organisasi ditekankan. Pemahaman tentang undang-undang berkaitan industri perakayuan di Malaysia dibincangkan.

This course covers the principles and activities in managing an organization. Management of human resource, production, marketing and finance as well as its relationship with the management of an organization are emphasized. An understanding of the laws related to the timber industry in Malaysia is discussed.

FHH4201 Teknologi Penglitup dan Kemasiapan Kayu / Wood Coating and Finishing Technology 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi pelbagai jenis penglitup dan laminasi permukaan produk kayu. Pemilihan teknik salutan mengikut jenis produk dan peralatan yang digunakan dalam proses kemasiapan ditumpukan. Kecacatan kemasan dan cara pemulihan dibincangkan.

This course encompasses various types of coating and surface lamination of wood products. Selection of coating technique according to types of wood products and equipment used in finishing process are emphasized. Finishing defects and remedial measures are discussed.

FHH4202 Penambahbaikan Mutu Kayu / Wood Quality Improvement 3 (2+1)

Prasyarat: Lulus / Pass FHH3103 dan/and FHH3206

Kursus ini merangkumi kelemahan kayu sebagai bahan mentah dan faktor yang mempengaruhinya. Prinsip dan kaedah rawatan untuk meningkatkan mutu kayu, serta penilaian prestasi ke atas kayu terawat ditumpukan. Perkembangan terkini dalam peningkatan mutu kayu dibincangkan.

This course encompasses the disadvantages of wood as raw material and factors which influence its quality. Principles and methods of treatments to enhance wood quality, and analyse the performance of the treated wood are emphasized. Latest developments on wood quality enhancement are discussed.

FHH4203 Biokomposit Termaju / Advanced Biocomposite 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi pemilihan bahan asas dan teknik pemprosesan untuk menghasilkan pelbagai panel biokomposit termaju berdasarkan aplikasi akhir yang bersesuaian. Pelbagai teknik untuk menghasilkan panel biokomposit termaju didedahkan. Faktor mempengaruhi sifat fizikal dan kekuatan mekanikal biokomposit termaju dibincangkan.

This course covers the selection of basic materials and processing techniques to produce a variety of advanced biocomposite panels based on appropriate end applications. Various techniques for producing advanced biocomposite panels are exposed. Factors influencing the physical properties and mechanical strength of advanced biocomposites are discussed.

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FHH4949 Projek Bacelor / Bachelor Project**6 (0+6)**

Prasyarat: Tiada

Kursus ini merangkumi aspek perancangan dan pelaksanaan projek. Ini termasuklah pemilihan tajuk, melaksanakan kajian kritis, mereka bentuk dan melaksanakan strategi kerja, mengumpul dan menganalisis data, serta mendokumentasi dan membentangkan hasil projek.

This course covers the aspects of planning and executing project. These include title selection, conducting critical review, designing and performing work strategy, collecting and analysing data, documenting and presenting project output.

FHH4959 Disertasi Bacelor / Bachelor Dissertation**6 (0+6)**

Prasyarat: Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.

JABATAN REKREASI DAN EKOPELANCONGAN /
DEPARTMENT RECREATION AND ECOTOURISM

FHR3100 Pengenalan Kepada Rekreasi Luar / *Introduction to Outdoor Recreation*

3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi konsep asas kesenggangan, rekreasi luar dan ekopelancongan. Hubungan antara sumber, pengguna dan pengurusan rekreasi dibincangkan. Skop, pembangunan dan profesionalisme industri rekreasi dan ekopelancongan ditekankan.

This course covers the basic concepts of leisure, outdoor recreation and ecotourism. The relationship between recreation resource, consumer and management are discussed. Scope, development and professionalisme in the recreation and ecotourism industry are emphasised.

FHR3112 Analisis Dalam Penyelidikan Rekreasi dan Ekopelancongan / *Research Analysis in Recreation and Ecotourism*

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi pemerihalan dan analisis kuantitatif dan kualitatif dalam bidang rekreasi dan ekopelancongan. Teknik pengukuran dan analisis data dipraktikan. Penekanan diberikan kepada interpretasi keputusan dan membuat rumusan.

This course covers the description and analysis of parametric and non -parametric data in recreation and ecotourism. Research design, methodology and data analysis are discussed. Emphasis is given to data analysis and interpretation.

FHR3202 Inventori Sumber Taman Dan Rekreasi / *Park and Recreation Rescouces Inventory*

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan kaedah inventori sumber semula jadi. Penggunaan peralatan untuk inventori dibincangkan. Penggunaan teknik inventori, pengumpulan data dan menganalisis sumber taman dan rekreasi ditekankan.

This course covers principles and methods of natural resources inventory. Utilisation of equipment for inventory is discussed. Application of inventory techniques, data collection and analysis of parks and recreation resource are emphasised.

FHR3203 Interpretasi Persekitaran / *Environmental Interpretation*

3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi teori dalam mengaplikasikan program interpretasi persekitaran. Demonstrasi teknik interpretasi terhadap kumpulan sasaran tertentu dibincangkan. Keberkesanan kaedah penyampaian interpretasi persekitaran melalui kemahiran komunikasi sehala ditekankan

This course covers the theory in the application of environmental interpretation program. Demonstration of interpretational techniques towards specific target groups are discussed. The effectiveness of the delivery methods of environmental interpretation through one-way communication skills are emphasized

FHR3204 Perancangan Rekreasi Luar / Outdoor Recreation Planning**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep perancangan kawasan rekreasi dalam menyediakan peluang rekreasi. Data sumber untuk pembangunan pelan perancangan kawasan rekreasi dianalisis. Perancangan pembangunan wilayah rekreasi dan ekopelancongan dibincangkan.

This course covers the concept of recreational area planning in providing recreational opportunities. Natural resource data for the development of recreation area planning plans are analyzed. Regional development planning for recreation and ecotourism is discussed.

FHR3208 Kemahiran Rekreasi Luar / Outdoor Recreation Skills**3 (0+3)**

Prasyarat: Tiada

Kursus ini merangkumi teknik asas dalam mengendalikan aktiviti rekreasi luar. Perancangan dan pengurusan pengendalian aktiviti rekreasi luar dibincangkan. Keupayaan mengendalikan pelbagai situasi sewaktu pelaksanaan aktiviti rekreasi luar ditekankan.

This course covers the basic technique in handling outdoor recreation activities. Planning and management of outdoor recreational skills activities handling is discussed. The ability to handle various situations during the implementation of outdoor recreational activities is emphasized.

FHR3301 Pengurusan Risiko Dalam Rekreasi / Risk Management in Recreation**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip pengurusan risiko bagi aktiviti rekreasi yang dikendalikan dalam persekitaran semulajadi. Kemahiran pengurusan risiko dan pengendalian situasi kecemasan dipraktikkan. Kepentingan pelan pengurusan risiko dalam pengendalian aktiviti rekreasi ditekankan.

This course covers the principles of risk management for recreational activities conducted in a natural environment. Risk management skills and handling emergency situations are practiced. The importance of a risk management plan in the operation of recreational activities is emphasized.

FHR3302 Pemrograman Dan Kepemimpinan Rekreasi / Recreation Programming and Leadership**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknik kepemimpinan dalam merangka program rekreasi. Langkah dan teknik pemrograman aktiviti rekreasi dibincangkan. Pembangunan dan pelaksanaan program rekreasi bagi pelbagai kumpulan pengguna rekreasi ditekankan.

This course covers the concept and technique of leadership in designing recreation program. Programming techniques and approaches of recreation activities are discussed. Development and implementation of recreation programs for various recreational user groups are emphasised.

FHR3501 Perilaku Pengguna Rekreasi / Recreational User Behaviour**3 (3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep asas perilaku pengguna dalam rekreasi. Kepelbagaian perilaku pengguna di kawasan taman dan rekreasi diperhatikan. Masalah perilaku pengguna di kawasan taman dan rekreasi dibincangkan.

This course covers the basic concept of consumer behaviour in recreation. The diversity of user behavior in park and recreation areas is observed. Consumer behaviour problems in parks and recreation are discussed.

- FHR3502 Perancangan dan Reka Bentuk Tapak Kawasan Rekreasi / Recreation Site Planning and Design 3 (2+1)**
- Prasyarat: Tiada
- Kursus ini merangkumi konsep reka bentuk kemudahan rekreasi di dalam persekitaran semula jadi. Langkah merancang pelan tapak rekreasi dan reka bentuk kemudahan rekreasi ditekankan. Pembentangan pelan kepada pelanggan dilaksanakan.
- This course encompasses the conceptual design of recreational facilities in a natural environment. The process of planning a recreation area and design of recreational are emphasized. Presentation of plan to client is implemented.*
- FHR3505 Pengurusan Hidupan Liar dalam Rekreasi / Wildlife Management in Recreation 3 (2+1)**
- Prasyarat: Tiada
- Kursus ini merangkumi konsep pengurusan hidupan liar dalam aktiviti rekreasi dan ekopelancongan. Pembangunan cadangan pengurusan konflik antara aktiviti rekreasi dan konservasi hidupan liar didedahkan. Peluang rekreasi yang bersesuaian dengan amalan konservasi hidupan liar dibincangkan.
- This course covers the concept of wildlife management in recreational and ecotourism activities. The development of conflict management proposals between recreational activities and wildlife conservation is revealed. Suitable recreational opportunities with wildlife conservation practices are discussed.*
- FHR3506 Rekreasi Terapeutik bagi Golongan Istimewa / Therapeutic Recreation for Special People 3 (2+1)**
- Prasyarat: Tiada
- Kursus ini merangkumi prinsip terapeutik dalam pemrograman rekreasi bagi golongan istimewa. Keperluan kemahiran khusus pengendalian program rekreasi terapeutik dipraktikkan. Aplikasi bagi kaedah yang bersesuaian dengan pelbagai golongan istimewa ditekankan.
- This course covers therapeutic principles in recreation programming for special group. The need for special skills in conducting therapeutic recreation programme is practiced. The application of appropriate methods to various special groups is emphasized.*
- FHR3507 Hospitaliti Dalam Taman dan Rekreasi / Hospitality in Parks and Recreation 3 (2+1)**
- Prasyarat: Tiada
- Kursus ini merangkumi konsep hospitaliti di dalam persekitaran rekreasi. Kualiti perkhidmatan kepada pelbagai kumpulan pengguna rekreasi dibincangkan. Pelbagai prosedur dan aktiviti perkhidmatan hospitaliti ditekankan.
- This course covers the concept of hospitality in recreational setting. Service quality towards various recreational user is discussed. Various hospitality service procedures and activities are emphasised.*
- FHR3509 Keusahawanan Rekreasi / Entrepreneurship in Recreation 3 (2+1)**
- Prasyarat: Tiada
- Kursus ini merangkumi kepelbagaian peluang perniagaan dalam bidang pengurusan taman dan rekreasi. Penyediaan rancangan perniagaan berdasarkan kehendak semasa pengguna ditekankan. Projek perniagaan berdasarkan pembangunan rancangan perniagaan dilaksanakan.
- This course discusses the diversity of business opportunities in parks and recreation management. Preparation of business plan according to the current user demands is emphasized. Business project based on developed business plan is implemented.*

FHR3510 Operasi dan Senggaraan Kemudahan Rekreasi / Operation and Maintenance of Recreational Facilities 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi prinsip operasi, teknik senggaraan, proses dan kaedah operasi, dan penyelenggaraan kemudahan rekreasi melalui perancangan. Pendekatan dalam membina plan operasi dan senggaraan kemudahan rekreasi didedahkan. Penilaian aktiviti pengoperasian dan senggaraan kemudahan rekreasi dibincangkan.

This course covers operational principles, maintenance techniques, operational processes and methods, and maintenance of recreational facilities through planning. The approach in developing an operation and maintenance plan for recreational facilities is exposed. The evaluation of recreational facility operation and maintenance activities is discussed.

FHR3901 Penjelajahan Amalan Terbaik Dalam Rekreasi Dan Ekopelancongan / Exploration of Best Practices in Recreation and Ecotourism 2 (0+2)

Prasyarat: Tiada

Kursus ini merangkumi perancangan, pentadbiran, dan pengurusan aktiviti penjelajahan rekreasi dan ekopelancongan. Kemahiran mengadaptasi dengan kepelbagaian budaya tempatan diterapkan. Konsep amalan terbaik dalam rekreasi dan ekopelancongan melalui penjelajahan ke destinasi serantau dibincangkan.

This course covers planning, administration, and management of recreational and ecotourism activities through exploration. The skills of adaptation to the diversity of local culture are applied. The concept of best practice in recreation and ecotourism through exploration to regional destinations are discussed.

FHR4102 Ekologi Sumber Rekreasi / Recreation Resource Ecology 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi komponen ekologi sumber rekreasi dalam pengurusan sumber secara mampan. Kepentingan pengukuran kesan impak rekreasi ke atas ekosistem sumber rekreasi dibincangkan. Hubungan kait kaedah pengurusan impak di kawasan rekreasi berdasarkan pola penggunaan sumber juga ditekankan.

This course covers the component of recreation resource ecology in sustainable resource management. The importance of impact measurement towards the recreational resources ecosystem is discussed. Relationship between impact management methods based on pattern of resource use are emphasised.

FHR4301 Penilaian Sumber Rekreasi / Recreation Resource Valuation 4 (4+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip ekonomi di dalam penilaian faedah dan kos dalam bidang rekreasi dan ekopelancongan. Kesejahteraan individu dan masyarakat serta kemampanan alam semulajadi dibincangkan. Teknik penilaian sumber rekreasi dan ekopelancongan dalam pembuatan keputusan terhadap penggunaan sumber ditekankan.

This course covers economic principles in the evaluation of benefits and costs in the field of recreation and ecotourism. Individual and community well-being and the sustainability of nature are discussed. Resource evaluation techniques for recreation and ecotourism in decision-making on resource use are emphasized.

FHR4302 Kaedah Penyelidikan Dalam Rekreasi Dan Ekopelancongan / Research Method in Recreation and Ecotourism 3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi kaedah penyelidikan dalam menyelesaikan masalah di dalam bidang rekreasi dan ekopelancongan. Pengajaran menekankan teknik-teknik persampelan, pengukuran dan prosedur statistik yang sesuai di dalam menyelesaikan masalah. Penentuan pendekatan rekabentuk penyelidikan yang sesuai di dalam sektor rekreasi dan ekopelancongan dibincangkan.

This course covers research methods in solving problems in recreation and ecotourism field. The teaching emphasises on sampling techniques, measurements and suitable statistical procedure in problem solving. Determination of suitable approaches of research design in recreation and ecotourism is discussed.

FHR4402 Pengurusan Pelancongan Warisan dan Komuniti / Heritage and Community Tourism Management 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi hubungan aspek warisan dan komuniti dalam konteks industri pelancongan. Pembangunan produk pelancongan warisan berasaskan komuniti dibincangkan. Kepentingan memelihara sumber warisan untuk tujuan industri pelancongan ditekankan.

This course covers the relationship of heritage and community aspects within the context of tourism industry. Development of community-based heritage tourism products is discussed. The importance of preserving heritage resources for tourism purposes are emphasised.

FHR4408 Pentadbiran dan Organisasi Taman dan Rekreasi / Administration and Organisation of Park and Recreation 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan amalan pentadbiran dan organisasi yang efektif di dalam bidang taman dan rekreasi. Amalan profesionalisme dalam pentadbiran dan pengurusan organisasi taman dan rekreasi dipraktikkan. Hubungan kait pentadbiran dan organisasi di dalam sektor awam dan swasta dibincangkan.

This course covers the principles and practices of effective administration and organization in the field of parks and recreation. The practice of professionalism in the administration and management of park and recreation organizations is practiced. The relationship between administration and organization in the public and private sectors is discussed.

FHR4501 Pendidikan Alam Semulajadi / Environmental Education 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi perkaitan pendidikan alam semulajadi dengan masalah persekitaran yang dihadapi oleh masyarakat. Program pendidikan alam semulajadi dirancang dan dilaksanakan. Kesedaran masyarakat terhadap pendidikan alam semulajadi ditekankan.

This course covers the relationship between environmental education and environmental problems faced by society. Environmental educational programs are planned and implemented. Community awareness towards environmental education is emphasised.

FHR4502 Pemasaran Rekreasi / Recreation Marketing 3 (2+1)

Prasyarat: Tiada

Kursus ini merangkumi konsep dan teknik pemasaran produk rekreasi. Strategi pemasaran untuk produk rekreasi dan ekopelancongan dibincangkan. Kaedah promosi untuk produk rekreasi dan ekopelancongan ditekankan.

This course covers the concepts and techniques of marketing recreation products. The marketing strategy for recreation and ecotourism products are discussed. Promotional methods for recreation and ecotourism products are emphasised

FHR4503 Pengurusan Ekopelancongan / Ecotourism Management**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep pengurusan ekopelancongan secara komprehensif. Aktiviti pengumpulan maklumat untuk pembangunan pelan ekopelancongan dilaksanakan. Cadangan pembangunan pelan pengurusan ekopelancongan ditekankan.

This course covers the concept of ecotourism management comprehensively. Gathering information activities for the development of an ecotourism plan are performed. Proposal for the development of an ecotourism management plan are emphasized.

FHR4504 Pembangunan Produk Rekreasi / Development of Recreation Products**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep pengguna, sumber, dan agensi dalam pembangunan dan pengkomersialan produk berasaskan perkhidmatan. Kaedah pengumpulan maklumat berkaitan pembangunan produk baharu diaplikasikan. Isu semasa dan pengaruh teknologi digital dalam membangunkan produk rekreasi dan ekopelancongan dibincangkan.

This course covers the concept of consumer behaviour, resources, and agencies in the development and product commercialization in services. Methods for gathering information related to a new product development are applied. Current issues and the influence of digital technology in product development is discussed.

FHR4608 Pengurusan Kawasan Perlindungan / Protected Areas Management**3 (2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan amalan terbaik pengurusan kawasan perlindungan. Simulasi pengurusan dan pengendalian konflik di kawasan perlindungan didedahkan. Menghasilkan cadangan pelan pengurusan dan pemantauan kawasan perlindungan ditekankan.

This course covers the concepts and best practices of conservation management. A simulation of management and conflict in protected areas is revealed. Proposing management plan and protected area monitoring are emphasized.

FHR4904 Kem Perhutanan / Forestry Camp**2 (0+2)**

Prasyarat: Tiada

Kursus ini mendedahkan kepada pelajar berkenaan kemahiran perhutanan dan kelangsungan hidup dalam persekitaran sebenar. Kemampuan menggunakan sumber yang terhad di kawasan sekitar dengan betul dan selamat ditekankan. Kemahiran pengurusan diri dan kumpulan diterapkan.

This course exposes students to forestry and survival skills in real environments. The ability to use limited resources in the environment correctly and safely is emphasized. Self- and group management skills are applied.

FHR4959 Disertasi Bachelo / Bachelor Dissertaion**6 (0+6)**

Prasyarat: Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.

JABATAN PENGURUSAN ALAM SEKITAR /
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT

EMP3101 Pengenalan Kepada Sistem Semulajadi / Introduction To Natural Systems

2(2+0)

Prasyarat: Tiada

Kursus ini merangkumi konsep asas sains alam sekitar dengan fokus kepada kesan dan penyelesaian terhadap isu perubahan iklim. Kursus ini mengkaji ekosistem daratan dan akuatik serta interaksi organisma dengan persekitaran. Selain itu, kursus ini menekankan proses ekologi utama dan impak perubahan iklim terhadap kestabilan ekosistem serta keseimbangan biodiversiti.

This course covers the fundamental concepts of environmental science with a focus on the impacts and solutions related to climate change. This course examines terrestrial and aquatic ecosystems as well as the interactions between organisms and their environment. In addition, the course emphasizes key ecological processes and the impact of climate change on ecosystem stability and biodiversity balance.

EMP3102 Pengurusan Kualiti Alam Sekitar / Environmental Quality Management

3 (3+0)

Prasyarat: Tiada

Kursus ini merangkumi aspek pengurusan kualiti alam sekitar dan kawalan pencemaran bandar. Elemen kesihatan dan risiko alam sekitar ditumpukan. Pemantauan dan teknologi kawalan yang bersesuaian bagi mengurangkan kesan pencemaran bagi memberikan persekitaran komuniti yang mampan turut ditekankan.

This course covers the aspects of environmental quality management and urban pollution control. Environmental health and risk elements are highlighted. Appropriate monitoring and control technology to reduce the impacts of the pollution in order to provide sustainable community environment are also emphasized.

EMP3103 Pengurusan Sumber Alam / Natural Resource Management

2(2+0)

Prasyarat: Tiada

Kursus ini merangkumi analisis kritikal terhadap isu eksploitasi sumber alam serta impaknya terhadap ekosistem global. Strategi pengurusan sumber yang berkesan bagi memastikan keseimbangan antara pemuliharaan hutan, hidupan liar, perikanan, air, dan tenaga dengan matlamat pembangunan mampan akan ditekankan. Cadangan pelan tindakan yang sistematik dan berkesan untuk setiap strategi pengurusan dalam usaha menangani masalah eksploitasi sumber alam juga dibincangkan.

This course covers a critical analysis of natural resource exploitation issues and their impacts on the global ecosystem. Emphasis is placed on effective resource management strategies to ensure a balance between the conservation of forests, wildlife, fisheries, water, and energy, and the goals of sustainable development. The development of systematic and effective action plans for each management strategy to address natural resource exploitation is also discussed.

EMP3104 Lukisan Teknikal Pelan dan Isometrik dalam Pengurusan Alam Sekitar / Plan and Isometric Technical Drawing in Environmental Management

2(0+2)

Prasyarat: Tiada

Kursus ini merangkumi konsep asas yang boleh diaplikasi dalam menguruskan alam sekitar. Teknik lakaran berkonsepkan 2D dan 3D dalam menggambarkan keadaan sesuatu persekitaran didedahkan. Pembangunan pelan berkenaan gambaran keadaan semasa persekitaran serta jangkaan perubahan keadaan persekitaran di masa hadapan ditekankan.

This course covers the basic concept that can be applied in managing the environment. Drawing techniques in the concept of 2D and 3D in describing the environment situation are exposed. The development of a plan that reflects the current situation of the environment and the expectation of future environmental changes is emphasized.

EMP3105 Kaedah Kuantitatif untuk Pengurusan Alam Sekitar / Quantitative Methods for Environmental Management 3(1+2)

Prasyarat: Tiada

Kursus ini merangkumi kaedah kuantitatif untuk menganalisis data dalam pengurusan alam sekitar, dengan penekanan terhadap teori dan amali asas matematik serta statistik. Pelajar akan melaksanakan pengumpulan dan analisis data kuantitatif berkaitan isu alam sekitar serta menggunakan perisian digital untuk memvisualisasikan kebolehubahan sistem iklim. Kursus ini juga membina kemahiran berangka dan kepimpinan pelajar dalam merancang penyelesaian berasaskan data bagi menyokong tindakan iklim mampan

This course covers quantitative methods for analyzing environmental management data, focusing on the fundamentals of mathematics and statistics theory and practice. Students will collect and analyze quantitative data related to environmental issues, using digital software to visualize climate system variability. The course also develops students' numerical and leadership skills in planning data-driven solutions to support sustainable climate action.

EMP3106 Sistem Maklumat Alam Sekitar / Environmental Information System 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi pengenalan sumber maklumat alam sekitar, kaedah pengumpulan dan pengurusan pangkalan data, mencari dan membentangkan maklumat. Pengurusan pangkalan data melalui penggunaan program komputer diperkenalkan

This course covers identification of sources of environmental information, methods of collecting, abstracting and presenting information, and database management. Database management using computers programs is also introduced.

EMP3107 Kemampanan dan Kemasyarakatan / Sustainability and Society 2(2+0)

Prasyarat: Tiada

Kursus ini merangkumi konsep dan prinsip pembangunan secara mampan. Perkaitan antara perancangan pembangunan fizikal dan manusia dihurai daripada perspektif sosio-budaya, politik dan ekonomi. Peranan tadbir urus dan pendidikan diuraikan.

This course covers the concept and principles of sustainable development. Relationship between human and physical planning development are discussed from the socio-cultural, political and economic perspectives. The role of governance and education are elaborated.

EMP3108 Prinsip Pengurusan Sisa Pepejal / Principles of Solid Waste Management 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi elemen fungsian dan analisis yang terlibat dalam sistem pengurusan sisa pepejal yang berkait dengan amalan semasa di Malaysia. Amalan terbaik melalui pengurusan sisa pepejal bersepadu bagi memperoleh sistem pengurusan yang lebih efisien dan mampan ditekankan.

This course covers the functional elements and analysis involved in solid waste management systems related to current practices in Malaysia. Best practices through integrated solid waste management are emphasized to achieve a more efficient and sustainable management system.

EMP3201 Pengenalan Perancangan Alam Sekitar / Introduction to Environmental Planning 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas dan teori asas perancangan bandar serta pelbagai kaedah perancangan bandar terkini. Aspek perancangan bandar fizikal seperti kajian pelan induk, pelan tempatan, pelan strategik dan penggunaan piawaian perancangan diperkenalkan. Hubungan kait antara perancangan, pengurusan, pemuliharaan alam sekitar bandar dan Sistem Maklumat Geografi ditekankan.

This course covers the fundamental principles and basic theories of urban planning, along with various current urban planning methods. Aspects of physical urban planning such as master plans, local plans, strategic plans, and the use of planning standards are introduced. The interconnection between planning, management, urban environmental conservation, and Geographic Information Systems (GIS) is emphasized.

EMP3202 Studio Perancangan Alam Sekitar I / Environmental Planning Studio I 3(0+3)

Prasyarat: Tiada

Kursus ini merangkumi konsep asas perancangan alam sekitar bandar dan pendekatan kerja studio untuk menghasilkan dan menerangkan pelan perancangan projek yang praktikal dan mampan. Kaedah penderiaan jarak jauh dan perisian pemetaan dalam menganalisis isu-isu perancangan bandar dan mencadangkan penyelesaian yang bersifat sistematik, strategik, dan jangka panjang ditekankan.

This course covers the basic concepts of urban environmental planning and a studio-based approach to produce and present practical and sustainable project planning designs. Remote sensing methods and mapping software are emphasized in analyzing urban planning issues and proposing systematic, strategic, and long-term solutions.

EMP3203 Penilaian Kesan Alam Sekitar dan Pelan Pengurusan Alam Sekitar / Environmental Impact Assessment and Environmental Management Plan 3(3+0)

Prasyarat: Lulus / Pass EMP3501 atau/or EMP3201

Kursus ini merangkumi pelaksanaan kaedah Penilaian Kesan Alam Sekitar (EIA) di dalam perancangan sesuatu projek atau tindakan. Penggunaan garis panduan pelaksanaan EIA bagi mengenal pasti, menganalisis serta melaporkan kekuatan dan kelemahan sesuatu projek didedahkan. Kaedah pelaporan berkaitan dengan tindakan mitigasi dan keperluan lain dalam mengurangkan impak kepada alam sekitar serta mewujudkan pembangunan bandar mampan melalui laporan Pelan Pengurusan Alam Sekitar (EMP) ditekankan.

This course covers the implementation of the Environmental Impact Assessment (EIA) method in the planning of projects or actions. The adaptation of EIA implementation guidelines to identify, analyze and report the strengths and weaknesses of a project is exposed. Reporting methods related to mitigation actions and other requirements in reducing environmental impact as well as creating sustainable urban development through the Environmental Management Plan (EMP) report are emphasized.

EMP3204 Geografi Fizikal / Physical Geography 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi isu penting dalam geografi fizikal dan menganalisis adaptasi yang diamalkan dalam pengurusan perubahan iklim. Interaksi global yang melibatkan ekosistem atmosfera dan lautan ditumpukan. Pengaruh proses global dan perubahan alam sekitar ke atas keadaan bentuk muka bumi, iklim dan langkah mitigasi pelepasan gas rumah hijau ditekankan.

This course covers the pertinent issues in physical geography and analyses adaptations practiced in climate change management. Global interactions involving the atmosphere and oceanic ecosystems are highlighted. The influence of global processes and environmental changes on landforms, climate and mitigation measures of green house gases are emphasised.

EMP3205 Pengurusan Kualiti Air dan Air Sisa / Water and Wastewater Quality Management 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi pengetahuan asas kualiti air dan piawaian yang berkaitan untuk air minum, air sisa dan pelepasan ke alam sekitar dengan cara yang mampan. Konsep pengurusan bekalan dan rawatan air mentah, rawatan air sisa dan penggunaan semula diserlahkan. Konsep pengurusan air secara bersepadu dititikberatkan.

This course covers the basic knowledge of water quality and related standards for drinking water, wastewater and discharge to the environment in a sustainable manner. The concept of raw water supply and treatment management, wastewater treatment and reuse are highlighted. The concept of integrated water management is emphasised.

EMP3206 Pengurusan dan Kawalan Pencemaran Udara dan Bunyi / Air and Noise Pollution Control and Management 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi pembelajaran berkenaan konsep asas tentang pencemaran udara dan bunyi. Polisi pengawalaturan piawai serta teknik pengukuran bagi cerapan data udara dan bunyi akan dedahkan kepada pelajar bagi tujuan menganalisis tahap pencemaran udara dan bunyi bagi sesuatu kawasan yang spesifik. Pelan tindakan strategik bagi pengurusan dan kawalan udara dan bunyi juga ditekankan bagi menyelesaikan masalah berkenaan pencemaran udara dan bunyi.

This course covers the learning about basic concepts on air and noise pollution. Standard regulatory policies and measurement techniques for air and noise data monitoring will be exposed to students for the purpose of analyzing the level of air pollution and noise for a specific area. A strategic action plan for the management and control of air and noise is also emphasized to solve the problem related to air and noise pollution.

EMP3207 Penilaian Kitar Hayat / Life Cycle Assessment 3(3+0)

Prasyarat: Tiada

Kursus Penilaian Kitar Hayat (LCA) ini merangkumi konsep dan metodologi yang digunakan untuk menilai kesan alam sekitar sesuatu produk atau proses dari awal hingga akhir hayat. Kursus ini menekankan tentang langkah-langkah analisis LCA, termasuk penetapan matlamat dan skop kajian, analisis inventori, penilaian impak alam sekitar, dan interpretasi keputusan kajian LCA. Kursus ini juga akan menekankan kepentingan LCA dalam pembangunan mampan dan mengurangkan kesan alam sekitar.

This Life Cycle Assessment (LCA) course covers important concepts and methodology used to assess the environmental impact of a product or process from the beginning to the end of life. This course emphasizes the steps of LCA analysis, including setting goals and scope of the study, inventory analysis, environmental impact assessment, and interpretation of LCA study results. This course will also emphasize the importance of LCA in sustainable development and reducing environmental impact.

EMP3208 Ekonomi Kitaran / Circular Economy 3(3+0)

Prasyarat: Tiada

Kursus ini meneroka pendekatan ekonomi kitaran, yang memfokuskan kepada pelbagai industri termasuk sektor kerajaan, separa kerajaan, dan swasta, bagi membolehkan pembangunan dan pengurusan operasi lebih berdaya tahan, berdaya saing dan mampan, seterusnya merangsang inisiatif kelestarian alam sekitar. Konsep keseluruhannya adalah berpaksikan kepada transisi daripada konsep "buaian ke pusara" ke "buaian ke buaian" ke arah agenda ekonomi hijau, serta mencapai matlamat pembangunan lestari. Isu dan cabaran, pemacu perubahan, pemangkin dan pemecut bagi ekonomi kitaran turut diketengahkan.

This course explores the circular economic approach, emphasising various parties, including government, semi-government, and private sectors, enabling the development of more resilient, competitive and sustainable operation and management, thereby fostering environmental sustainability. The overall concept is to transition from "cradle to grave" to "cradle to cradle" in an effort to advance the green economic agenda and achieve Sustainable Development Goal. Issues and challenges, change drivers, as well as catalysts and accelerators for the circular economy, are highlighted

EMP3209 Pengurusan Alam Sekitar dalam Perspektif Islam / *Environmental Management in Islamic Perspective* **2(1+1)**

Prasyarat: Tiada

Kursus ini memfokuskan pada pemahaman konsep Islam tentang pengurusan alam sekitar dan tanggungjawab manusia sebagai khalifah. Kursus ini mengaplikasikan prinsip-prinsip ini melalui penganjuran seminar yang membahaskan isu-isu alam sekitar dalam konteks Islam. Kursus ini juga memberi penekanan kepada adab dan etika yang diperlukan dalam pengurusan alam sekitar, sesuai dengan ajaran Islam, untuk memupuk sikap bertanggungjawab dan beretika dalam interaksi manusia dengan alam sekitar.

This course focuses on understanding the Islamic concept of environmental management and human responsibilities as caliphs. This course applies these principles through the organization of seminars that discuss environmental issues in the context of Islam. This course also emphasizes the manners and ethics required in environmental management, in accordance with Islamic teachings, to foster responsible and ethical attitudes in human interaction with the environment.

EMP3210 Aplikasi GIS Dan Penderiaan Jarak Jauh kepada Pengurusan Alam Sekitar / *GIS and Remote Sensing Applications to Environmental Management* **3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi pemahaman asas sistem GIS dan penderiaan jarak jauh dan aplikasinya dalam bidang berkaitan pengurusan alam sekitar. Asas geoinformatik dan manipulasi data geospasial turut ditekankan. Prinsip penderiaan jarak jauh dan kaedah asas pemprosesan digital penderiaan jarak jauh dalam sistem GIS untuk aplikasi dalam pengurusan alam sekitar juga dibincangkan.

This course covers the basic understanding of GIS and remote sensing systems and their applications in the field of environmental management. Fundamentals of geoinformatics and geospatial data manipulation are also emphasized. The principles of remote sensing and the basic methods of digital processing of remote sensing in GIS systems for applications in environmental management are also discussed.

EMP3301 Ekonomi dan Pengurusan Alam Sekitar / *Environmental Economics and Management* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi pendekatan ekonomi dalam mentafsir masalah alam sekitar dan menilai penyelesaian dasar. Kegagalan pasaran dan eksternaliti, serta penyelesaian lazim dan ekonomi untuk masalah alam sekitar ditumpukan. Pendekatan analisis risiko, analisis kos-faedah, kajian kes masalah alam sekitar, serta pendekatan untuk mencapai pembangunan bandar mampan turut diterapkan.

This course covers the economic approach in assessing environmental problems and evaluating policy solutions. Market failure and externalities, as well as conventional and economic solutions to environmental problems are highlighted. Risk analysis approaches, benefit-cost analysis, case studies in environmental problems, and approaches to achieve sustainable urban development are also covered.

EMP3302 Pengurusan dan Operasi Projek / *Project Management and Operation* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi teori, teknik dan proses perancangan serta pengurusan strategik projek di peringkat makro dan mikro, dengan tumpuan kepada elemen pengurusan dan pengawalan kualiti serta impak projek. Analisis projek, teknik numerasi penilaian projek dan isu kemampanan dan ketahanan sistemik dalam perancangan dan pembangunan projek diberi penekanan.

This course covers the theory, techniques and processes of planning and strategic project management at the macro and micro levels, with a focus on the element of management and quality control as well as project impact. Project analysis, numerical techniques for project evaluation and issues of sustainability and systemic resilience in project planning and development are emphasised.

EMP3303 Perbandingan Perancangan dan Pengurusan Alam Sekitar / Comparative Environmental Planning and Management

3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi penerokaan mendalam terhadap prinsip dan amalan perancangan alam sekitar merentasi pelbagai wilayah dan negara. Kursus ini menekankan pemahaman yang menyeluruh tentang persamaan, perbezaan, dan cabaran dalam pendekatan perancangan alam sekitar di peringkat global. Kursus ini merangkumi rangka kerja teori, kajian kes, dan aplikasi praktikal bagi membentuk perspektif holistik terhadap pembangunan mampan dan pengurusan alam sekitar.

This course covers an in-depth exploration of environmental planning principles and practices across various regions and countries. This course emphasizes a comprehensive understanding of the similarities, differences, and challenges in global environmental planning approaches. The course incorporates theoretical frameworks, case studies, and practical applications to foster a holistic perspective on sustainable development and environmental management.

EMP3401 Masyarakat, Etika dan Alam Sekitar / Society, Ethics and the Environment

3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi pertalian antara masyarakat dan isu alam sekitar masa kini. Peranan etika, kerajaan, undang-undang dan ekonomi dalam menyelesaikan masalah alam sekitar, serta penglibatan masyarakat dalam kawalan pencemaran, pengurusan sumber secara lestari, dan pembangunan kesedaran serta keupayaan untuk menangani dan menyesuaikan diri terhadap perubahan persekitaran secara mampan turut ditekankan.

This course covers the relationship between society and current environmental issues. It emphasizes the roles of ethics, government, law, and the economy in addressing environmental problems, as well as community involvement in pollution control, sustainable resource management, and the development of awareness and capacity to respond and adapt sustainably to environmental changes

EMP3402 Kaedah Penyelidikan Kualitatif untuk Alam Sekitar / Qualitative Research Methods for the Environment

4(3+1)

Prasyarat: Tiada

Kursus ini merangkumi pendekatan dan kaedah asas penyelidikan termasuk kerangka teori dalam penyelidikan kualitatif alam sekitar. Kaedah pengumpulan data termasuk temubual, pemerhatian dan analisis dokumen ditumpukan.

This course covers qualitative research methods in the field of environment including theoretical framework in qualitative research. Data collection methods such as interviews, observations and document analysis are highlighted.

EMP3501 Pengenalan kepada Undang-Undang Alam Sekitar / Introduction to Environmental Law

3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi pengenalan kepada sistem perundangan di Malaysia mengikut Perlembagaan Persekutuan dan relevan undang-undang lazim terhadap isu alam sekitar. Kursus ini memberi fokus kepada pelbagai undang-undang statut berkaitan kawalan pencemaran alam sekitar dalam membentuk norma masyarakat, selain kesedaran tentang kepentingan penguatkuasaan. Pembentukan undang-undang alam sekitar antarabangsa melalui perjanjian pelbagai-hala turut dibincangkan.

This course covers the pertinent issues in physical geography and analyses adaptations practiced in climate change management. Global interactions involving the atmosphere and oceanic ecosystems are highlighted. The influence of global processes and environmental changes on landforms, climate and mitigation measures of green house gases are emphasised.

EMP3801 Pengenalan kepada Kerja Lapangan Pengurusan Alam Sekitar / Introduction to Environmental Management Fieldwork 3(1+2)

Prasyarat: Tiada

Kursus ini memperkenalkan kaedah dan teknik pengumpulan serta analisis data sains dan sosial alam sekitar secara sistematik. Pelajar akan mengaplikasikan kaedah dan teknik yang beresuaian bagi setiap parameter yang hendak diukur. Kaedah pensampelan serta menganalisis data alam sekitar menggunakan teknologi digital secara efektif juga ditekankan. Kursus ini turut menekankan konsep saintifik, kecekapan teknikal, nilai kepimpinan, dan pelaksanaan projek komuniti dalam topik terpilih.

This course introduces systematic methods and techniques for collecting and analyzing environmental science and social data. Students will apply appropriate methods and techniques for measuring each parameter. The course also emphasizes on scientific concepts, technical competencies, leadership values, and the integration of community-based projects in selected topics.

EMP4101 Sistem Pengurusan Kualiti Alam Sekitar / Environmental Quality Management System 3(3+0)

Prasyarat: Tiada

Kursus ini memberi tumpuan kepada penerapan Sistem Pengurusan Alam Sekitar (EMS) dengan pendekatan pemikiran sistem untuk menyokong tindakan iklim yang mampan. Kursus ini menekankan analisis piawaian EMS dan menunjukkan kemahiran kepimpinan dalam melaksanakan perubahan sistematik yang berkesan dalam organisasi.

This course focuses on the application of Environmental Management Systems (EMS) using a systems thinking approach to support sustainable climate action. It emphasizes the analysis of EMS standards and demonstrates leadership skills in implementing effective systematic changes within organizations.

EMP4102 Statistik untuk Perancangan dan Pengurusan Alam Sekitar / Statistics for Environmental Planning and Management 3(2+1)

Prasyarat: Lulus / Pass EMP3105

Kursus ini merangkumi kaedah kuantitatif untuk menganalisis data dalam konteks pengurusan alam sekitar, dengan penekanan kepada teori dan amali asas matematik serta statistik. Pengumpulan dan analisis data kuantitatif berkaitan isu alam sekitar akan dilaksanakan, termasuk penggunaan perisian digital untuk memvisualisasikan kebolehubahan sistem iklim. Kursus ini turut membina kemahiran berangka dan kepimpinan dalam merancang penyelesaian berasaskan data bagi menyokong tindakan iklim yang mampan.

This course covers quantitative methods for data analysis in the context of environmental management, with emphasis on fundamental mathematical and statistical theories and applications. Quantitative data collection and analysis related to environmental issues will be carried out, including the use of digital software to visualize climate system variability. The course also develops numerical and leadership skills in designing data-driven solutions to support sustainable climate action

EMP4103 Pengurusan Projek Dalam Sekitaran Terkawal / Management of Projects in Controlled Environments 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi penggunaan dan aplikasi amalan terbaik dan pendekatan berasaskan proses untuk pengurusan pelbagai projek berkaitan alam sekitar. Kajian kes perniagaan, kumpulan pengurusan projek, kawalan, risiko dan kualiti ditumpukan.

This course covers the use and application of best practices and process-based approaches for the management of various environment-related projects. It focuses on business case studies, project management teams, control, risk, and quality.

EMP4104 Perubahan Alam Sekitar dan Isu Semasa / *Environmental Changes and Current Issues* **2(2+0)**

Prasyarat: Tiada

Kursus ini bertujuan untuk memberikan pemahaman menyeluruh mengenai perubahan alam sekitar dan isu semasa di pelbagai peringkat. Kursus ini melatih pelajar menganalisis secara kritis kesan aktiviti manusia terhadap kualiti alam sekitar serta mengkaji hubungan antara pembangunan dan degradasi ekosistem. Melalui pendekatan interdisiplin, mereka akan didorong untuk mencadangkan strategi inovatif dalam menangani isu-isu alam sekitar dan mengurangkan impak negatifnya.

This course aims to comprehensively understand environmental changes and current issues at various levels. The course trains the students to critically analyze the impact of human activities on environmental quality and examine the relationship between development and ecosystem degradation. Through an interdisciplinary approach, they will be encouraged to propose innovative strategies to address environmental issues and mitigate negative impacts.

EMP4105 Pengurusan Pesisir Laut dan Marin / *Coastal and Marine Management* **3(3+0)**

Prasyarat: Tiada

Kursus ini memberi tumpuan kepada pengurusan kawasan pesisir laut dan marin serta kesan pembangunan terhadap kawasan tersebut. Kursus ini merangkumi perbincangan tentang proses semula jadi dan antropogenik yang mempengaruhi kawasan ini, serta amalan terbaik pengurusan dan kajian kes berkaitan.

This course focuses on coastal and marine management, as well as the impact of development on these areas. The course includes discussions on both natural and anthropogenic processes that affect coastal and marine environments, along with best management practices and relevant case studies.

EMP4201 Studio Perancangan Alam Sekitar II / *Environmental Planning Studio II* **3(0+3)**

Prasyarat: Lulus / *Pass* EMP3201 dan/and EMP3202

Kursus ini merangkumi aspek strategik dan kepelbagaian disiplin yang diperlukan dalam perancangan alam sekitar yang luas termasuk Rancangan Pembangunan Negara (RPN) bagi pembentukan pembangunan bandar dan desa secara komprehensif atau sebagai kajian pengurusan sumber. Input saintifik, teknik penyelidikan, interaksi ekologi, sosial, ekonomi serta aspek budaya ditumpukan. Penggunaan kaedah penderiaan jauh dan perisian pemetaan ditekankan.

This course covers the strategic and multi-disciplinary nature of broad scale environmental planning including National Physical Plan (NPP) for shaping comprehensive urban and rural development or as a resource management study. Scientific inputs and research techniques, interaction of ecological, social, economic and cultural aspects are highlighted. Remote sensing methods and mapping software techniques are emphasized.

EMP4202 Perancangan Pengangkutan / *Transportation Planning* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi sistem pengangkutan dan hubungannya dengan guna tanah di kawasan bandar dan desa. Teori aliran trafik dan implikasi pembangunan dan kegunaan tanah bagi kemudahan pengangkutan ditumpukan. Pengumpulan data, teknik penilaian dan teknik penentuan projek serta perancangan pengangkutan di samping kesannya dibincangkan.

This course covers transport systems and its relationship to land use in urban and rural areas. Traffic flow theory and the implications of development and various land use activities on transport facilities are highlighted. Data collection, evaluation and project appraisal techniques for transport planning and its effects are discussed.

EMP4203 Perancangan dan Pengurusan Alam Sekitar Gunaan / *Applied Environmental Planning and Management* **3(2+1)**

Prasyarat: Lulus / Pass EMP3204

Kursus ini merangkumi konsep dan kaedah yang diamalkan dalam perancangan dan pengurusan alam sekitar. Kemahiran analisis untuk memilih pelan alternatif yang sesuai dan langkah-langkah penebatan untuk mengurangkan impak persekitaran ditekankan.

This course covers the concepts and methods practiced in environmental planning and management. Analytical skills in selecting appropriate alternative plans and mitigation measures to reduce environmental impacts are emphasized.

EMP4204 Perancangan Respons Tumpahan Minyak / *Oil Spill Response Planning* **2(2+0)**

Prasyarat: Tiada

Kursus ini merangkumi strategi dan isu yang terlibat dalam respons kepada insiden tumpahan minyak di laut. Fakta yang mempengaruhi perlakuan dan pergerakan minyak, serta impak terhadap jenis pantai dan ekosistem diuraikan. Komponen pelan kontingensi dan langkah yang terlibat dalam pengurusan respons tumpahan minyak dibincangkan.

This course covers the strategies and issues involved in response to oil spills at sea. The factors which influence the behavior and movement of oil, including the impact on different types of shores and ecosystems are explained. The components of an oil spill contingency plan and steps involved in the management of oil spill responses are discussed.

EMP4205 Alam Sekitar, Sosial dan Tadbir Urus / *Environment, Social and Governance* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi penerokaan menyeluruh tentang prinsip dan amalan yang berkaitan dengan alam sekitar, sosial dan tadbir urus (ESG). Kursus ini merangkumi komponen ESG, pelaporan ESG dan pengukuran prestasi. Potensi pelaksanaan ESG dalam pengurusan warisan semula jadi dan amalan perniagaan mampan turut diterokai.

This course covers a comprehensive exploration of the principles and practices related to Environmental, Social, and Governance (ESG). The course covers ESG components, ESG reporting, and performance measurement. The potential implementation of ESG in natural heritage management and sustainable business practices is also explored

EMP4301 Penilaian Perkhidmatan Ekosistem / *Ecosystem Services Valuation* **3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi pendekatan ekonomi dalam penilaian sumber asli dan perkhidmatan ekosistem. Pelbagai kaedah, strategi, teknik dan penggunaan perisian ekonomi untuk menilai sumber asli dan perkhidmatan ekosistem ditekankan. Pengukuran kebajikan, kajian kes yang berkaitan dengan masalah pengurusan sumber asli, dan juga pendekatan untuk mencapai kemampanan alam sekitar bagi penyelesaian berpanduan dasar perubahan iklim negara turut dibincangkan.

This course covers economic approaches in the assessment of natural resources and ecosystem services. Various methods, strategies, techniques, and the use of economic software for evaluating natural resources and ecosystem services are emphasized. Welfare measurement, case studies related to natural resource management issues, and approaches to achieving environmental sustainability in line with national climate change policy solutions are also discussed.

EMP4401 Geografi Manusia / Human Geography 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi pelbagai pendekatan geografi manusia khususnya dari aspek budaya, sosial, politik dan ekonomi. Kursus ini juga menekankan aspek geografi manusia di dalam konteks pembangunan mampan.

This course encompasses various approaches to human geography, particularly from the cultural, social, political, and economic aspects. The course also emphasizes the aspects of human geography in the context of sustainable development

EMP4402 Gender dan Pengurusan Sumber Asli Bersepadu / Gender and Integrated Natural Resources Management 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi isu gender dalam pengurusan sumber asli bersepadu. Pertimbangan aspek gender dalam projek bersepadu pengurusan sumber asli dan kelemahan projek yang tidak peka kepada aspek gender dibincangkan melalui kajian kes. Kaedah petunjuk kesan tindakan juga dibincangkan

This course covers the issue of gender in integrated natural resource management. Consideration of gender aspects in integrated natural resource management and weaknesses in projects which are insensitive to gender aspects are discussed through case studies. Indicators of impact methods are also discussed.

EMP4403 Penyelidikan Dalam Sosiologi Alam Sekitar / Research in Environmental Sociology 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi perhubungan antara alam sekitar dan masyarakat, serta mengaitkan teori sosial kepada penyelidikan alam sekitar. Kajian sosiologi yang memberi fokus kepada alam sekitar sebagai satu pendekatan kepada kemampanan ditumpukan.

This course covers the study of relations between environment and society and relate social theories to environmental research. Sociological research focusing on environmental matters as an approach to sustainability are highlighted

EMP4404 Eko-Pelancongan dan Alam Sekitar / Ecotourism and Environment 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi pembangunan inisiatif alam sekitar dan sosial berasaskan industri pelancongan. Perancangan strategik kemampanan di dalam memastikan entiti perniagaan ekopelancongan yang diusahakan diurus secara profesional dan berjaya mencapai matlamat ditekankan.

This course encompasses development of environmental and social initiatives in the tourism industry. Planning of sustainable strategies in ensuring the ecotourism business entity is managed professionally and successfully achieves the objectives are emphasized.

EMP4501 Polisi dan Pentadbiran Alam Sekitar / Environmental Policy and Administration 3(3+0)

Prasyarat: Lulus / Pass EMP3501

Kursus ini merangkumi proses pembentukan dan pelaksanaan polisi yang berkaitan dengan alam sekitar. Penekanan diberikan kepada pendekatan polisi dan pentadbiran alam sekitar di Malaysia. Kursus ini turut memberi pendedahan kepada polisi dan sistem pentadbiran alam sekitar yang diamalkan di negara lain.

This course covers the process of formulating and implementing environmental policies. Emphasis is given to policy approaches and environmental governance in Malaysia. The course also provides exposure to environmental policies and administrative systems practiced in other countries.

EMP4502 Undang-undang Perancangan dan Alam Sekitar / *Planning Law and Environment* 3(3+0)

Prasyarat: Lulus / Pass EMP3501

Kursus ini merangkumi peranan undang-undang perancangan bandar dan desa di Malaysia dan perkaitannya dengan perlindungan alam sekitar. Pelbagai isu dan cabaran pelaksanaan undang-undang perancangan bagi mengatasi masalah alam sekitar setempat akan dibincangkan. Penguatkuasaan oleh pelbagai peringkat pentadbiran kerajaan akan turut ditekankan.

This course covers the role of town and country planning law in Malaysia and its relation to environmental protection. Various issues and challenges of implementing the planning law to overcome local environmental problems will be discussed. Enforcement by various levels of government administration will also be emphasized.

EMP4503 Undang-Undang berkaitan Pemuliharaan Kepelbagaian Biologi dan Perubahan Iklim / *Laws relating to Biodiversity Conservation and Climate Change* 3(3+0)

Prasyarat: Lulus / Pass EMP3501

Kursus ini merangkumi undang-undang dan polisi berkaitan pemuliharaan kepelbagaian biologi dan perubahan iklim di Malaysia. Perjanjian pelbagai hala berkaitan turut dibincangkan. Kursus ini juga melihat kepada aspek keadilan alam sekitar daripada perspektif undang-undang, selain mendedahkan pelajar kepada kes-kes mahkamah dalam Malaysia dan luar negara.

This course comprises laws and policies related to biodiversity conservation and climate change in Malaysia. Related multilateral agreements are also discussed. This course also looks at aspects of environmental justice from a legal perspective, apart from exposing students to court cases in Malaysia and other jurisdictions.

EMP4801 Kerja Lapangan Bersepadu Pengurusan Perancangan Alam Sekitar / *Integrated Environmental Management Planning Fieldwork* 3(0+3)

Prasyarat: Tiada

Kursus ini merangkumi kaedah penyelidikan melalui kerja lapangan dengan penglibatan daripada komuniti setempat bagi menyelesaikan isu alam sekitar yang dihadapi. Teknik pengumpulan dan analisis data serta pelaporan projek kerja lapangan ditekankan.

This course covers research methods through fieldwork with the involvement of the local community to solve the encountered environmental issue. Techniques for data collection, analysis, and reporting of fieldwork projects are emphasized

EMP4902 Kaedah Penyelidikan untuk Pengurusan Alam Sekitar / *Research Methodologies for Environmental Management* 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi kaedah asas penyelidikan secara kualitatif dan kuantitatif untuk mengintegrasikan pendekatan penyelesaian masalah yang berkesan dalam pengurusan alam sekitar. Permasalahan kajian, hipotesis, dan objektif kajian serta sorotan kajian berdasarkan isu alam sekitar diperincikan. Aplikasi teori penyelidikan untuk pembentukan kertas cadangan penyelidikan pengurusan alam sekitar ditekankan dalam menyusun strategi penyelidikan yang lestari dan bertanggungjawab terhadap masyarakat dan alam sekitar.

This course covers basic qualitative and quantitative research methods to integrate effective problem-solving approaches in environmental management. The research problem, hypothesis, and research objectives as well as research highlights based on environmental issues are detailed. The application of research theory to the formation of environmental management research proposal papers is emphasized in formulating research strategies that are sustainable and responsible for society and the environment.

EMP4903 Latihan Industri / Industrial Training**12(0+12)**

Prasyarat: Telah melengkapkan dan lulus semua kursus termasuk pakej ELEX / *Have completed and passed all courses including the ELEX package*

Kursus ini merangkumi pendedahan kepada persekitaran kerja sebenar di industri/organisasi yang mampan. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan kemampanan secara bersepadu semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi bersepadu turut diberi penekanan.

This course covers exposure to real work environments in sustainable industries/organizations. The training includes the application of theoretical and practical aspects learned, with an emphasis on sustainable integrated collaboration in the workplace. Problem-solving and integrated communication skills are also emphasized

EMP4904 Teknik Seminar Pengurusan Alam Sekitar / Environmental Management Seminar Techniques**2(1+1)**

Prasyarat: Tiada

Kursus ini merangkumi teknik pembentangan seminar termasuk persembahan, isi, persiapan, alat pandang dengar dan perisian persembahan. Pemilihan isu semasa alam sekitar yang berkaitan dengan projek berasaskan komuniti ditumpukan. Pengurusan seminar isu semasa alam sekitar yang berkesan ditekankan.

This course covers seminar presentation techniques, including delivery, content, preparation, audiovisual tools, and presentation software. Emphasis is placed on the selection of relevant current environmental issues related to community-based project. Effective management of seminars on contemporary environmental topics is also highlighted.

EMP4905 Pemudahcaraan dalam Pengurusan Alam Sekitar / Facilitation in Environmental Management**2(1+1)**

Prasyarat: Tiada

Kursus ini merangkumi adaptasi teknik memudahcara untuk menilai isu-isu alam sekitar. Proses memudah cara diadaptasi untuk membincangkan isu alam sekitar

This course includes the adaptation of facilitation techniques to assess environmental issues. Facilitating processes are adapted to discuss environmental issues.

EMP4959 Disertasi Bacelor / Bachelor Dissertation**6(0+6)**

Prasyarat: Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai diberi penekanan, termasuk dalam meneroka isu pembangunan mampan bandar secara inklusif dan berimpak kepada masyarakat.

This course covers the preparation of proposals, implementation, and scientific writing for a research project. Emphasis is given to a scientific approach in systematically generating data through appropriate design, collection, and analysis, including the exploration of sustainable urban development issues that are inclusive and impactful to communities.

JABATAN SAINS DAN TEKNOLOGI ALAM SEKITAR /
DEPARTMENT OF ENVIRONMENTAL SCIENCE AND TECHNOLOGY

ESC3022 Sisa Pepejal dan Sisa Berbahaya / Solid Waste and Hazardous Waste

3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi perbezaan jenis dan sumber sisa pepejal serta sisa berbahaya, dengan penekanan kepada amalan pengurusan sisa yang mampan. Kaedah rawatan dan pelupusan sisa yang efektif melalui pendekatan minda keusahawanan untuk menyokong penyelesaian lestari di tekankan. Kursus ini turut menilai rangka kerja perundangan dan kempen penglibatan awam dengan tumpuan kepada kemampanan dan kesedaran sendiri sejajar dengan tanggungjawab individu dalam mencapai Matlamat Pembangunan Mampan (SDG 12).

This course covers the differences between the types and sources of solid and hazardous waste, with an emphasis on sustainable waste management practices. Effective methods of waste treatment and disposal through an entrepreneurial mindset to support sustainable solutions are emphasized. The course also evaluates the legal framework and public engagement campaigns, focusing on sustainability and self-awareness in fulfilling individual responsibilities aligned with Sustainable Development Goal (SDG 12)

ESC3023 Analisis dan Instrumentasi Alam Sekitar / Instrumentation and Environmental Analysis

3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi analisis kimia dalam alam sekitar termasuk analisis pencemar organik, pencemar tak organik dan reaktiviti bahan pencemar dengan partikel. Penyediaan sampel, bahan kimia piawai dan peralatan diperincikan. Instrumentasi terkini yang digunakan dalam kajian alam sekitar dibincangkan.

This course covers chemical analysis in the environment that includes analysis of organic and inorganic pollutants and particle reactivity with pollutants. Preparation of samples, chemical standards and equipment are detailed out. The most recent instrumentations employed in environmental studies are discussed

ESC3024 Teknologi Rawatan Sisa Pepejal / Solid Waste Treatment Technology

3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi elemen teknologi pengurusan sisa pepejal bermula daripada penjanaan sehingga ke pelupusan akhir. Hubungkait antara ekonomi, teknologi, kesihatan, alam sekitar, perundangan dan perancangan dalam sistem pengurusan sisa pepejal ditekankan.

This course covers solid waste management technology, from generation until final disposal. The relationship between economics, technologies, health, environment, law, and planning in solid waste management is emphasized.

ESC3025 Kemampanan, Etika dan Isu Alam Sekitar / Sustainability, Ethics and Environmental Issues

3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan matlamat berlandaskan Matlamat Pembangunan Mampan (SDG) dalam membincangkan isu alam sekitar di peringkat nasional dan antarabangsa. Perkongsian amalan kemampanan dalam kehidupan harian dibincangkan. Aspek etika alam sekitar dalam membincangkan isu alam sekitar dan dalam membuat keputusan berkaitan pembangunan mampan turut ditekankan.

This course covers the principles and goals based on the Sustainable Development Goals (SDGs) in discussing environmental issues at the national and international levels. Sharing of sustainability practices in daily life is discussed. Environmental ethics aspect in debating environmental issues and in making sustainable development decisions are also emphasized.

ESC3026 Ekologi Biosfera / Biosphere Ecology**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi pengenalan kepada komponen biosfera dan interaksi biotik-abiotik dalam alam sekitar. Kesan impak aktiviti manusia terhadap komponen biosfera dan peranan setiap komponen dalam mendepani perubahan alam sekitar dibincangkan. Hierarki kerumitan dalam ekologi biosfera dari peringkat spesies, populasi, komuniti, ekosistem sehingga pengurusannya di peringkat biosfera turut ditekankan.

This course covers the introduction to global ecological system and biotic-abiotic interactions in the environment. The impact of human activity on the biosphere component and each component's role in facing environmental change is discussed. The hierarchy of complexity in biosphere ecology from species, population, community, ecosystem to management at the biosphere level is emphasized.

ESC3027 Hidrosfera dan Litosfera / Hydrosphere and Lithosphere**3(3+0)**

Prasyarat: Tiada

Kursus ini memperkenalkan struktur dalaman bumi dan kitaran air semula jadi, dengan penekanan terhadap interaksi antara hidrosfera dan litosfera. Ia turut menekankan kepentingan melindungi dan memulihkan ekosistem berkaitan air seperti sungai, tanah lembap, dan akuifer dalam konteks pengurusan alam sekitar secara mampan.

This course introduces Earth's internal structure and water cycle, emphasizing hydrosphere-lithosphere interactions. It also highlights the importance of protecting and restoring water-related ecosystems such as rivers, wetlands, and aquifers, within the context of sustainable environmental management.

ESC3028 Teknik Lapangan Alam Sekitar / Environmental Fieldwork Techniques**2(0+2)**

Prasyarat: Tiada

Kursus ini memperkenalkan prinsip asas dan kaedah pengukuran parameter alam sekitar merangkumi aspek air, tanah, biota, kesihatan, sisa dan kajian sosial. Teknik pengukuran, persampelan, analisis dan interpretasi data serta pembentangan laporan turut ditekankan bagi memelihara alam sekitar.

This course introduces the basic principles and methods of measuring environmental parameters which include aspects of water, soil, biota, health and waste. The techniques of measurement, sampling, analysis and interpretation of data as well as report presentation are also emphasized to preserve the environment.

ESC3029 Sains Atmosfera dan Perubahan Iklim / Atmospheric Science and Climate Change**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi pengenalan kepada konsep asas berkenaan komposisi dan struktur atmosfera serta proses fizikal dan proses dinamik yang menyebabkan perubahan cuaca dan iklim. Senario perubahan iklim semasa, pelepasan gas rumah hijau, impak perubahan iklim, dan strategi mitigasi dan adaptasi selaras dengan Matlamat Pembangunan Mampan (SDG) 13.3 turut dibincangkan.

This course covers introduction to the fundamental concepts of atmospheric compositions and structures as well as physical and dynamical processes leading to weather and climate change. Current climate change scenarios, greenhouse gas emissions, impact of climate change, and mitigation and adaptation strategies aligned with Sustainable Development Goal (SDG) 13.3 are also discussed.

ESC3102 Prinsip Kesihatan Alam Sekitar / Principles of Environmental Health**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi pengenalan kepada hubungan kompleks di antara kesihatan umum dan persekitaran. Isu dan masalah persekitaran yang memberi kesan ke atas kesihatan manusia berdasarkan kaedah toksikologi serta penilaian risiko yang mampan juga dibincangkan. Penilaian prinsip kesihatan persekitaran dan kesihatan di tempat kerja turut ditekankan.

This course covers the introduction to the complex relationship between public health and the environment. Issues and environmental problems affecting human health based on toxicology and risk assessment methods are also discussed. Evaluation of environmental health principles and workplace health are also emphasized.

ESC3206 Hidrologi Alam Sekitar / Environmental Hydrology**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kaedah hidrologi dalam perancangan dan pengurusan alam sekitar. Kaedah organisasi data hidrologi bagi pengurusan lembangan secara mampan serta kesan aktiviti manusia terhadap persekitaran hidrologi turut dibincangkan.

This course covers the hydrological methods in environmental planning and management. Organization of hydrological data for sustainable watershed management and human impacts on environmental hydrology are also discussed.

ESC3207 Rawatan Air Sisa / Wastewater Treatment**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi proses rawatan air sisa berasaskan kepada keperluan perundangan tempatan, sumber dan ciri air sisa dan peranan loji rawatan. Parameter asas dalam menguji keberkesanan operasi loji, pengawasan proses serta isu kesihatan dan keselamatan loji ditekankan. Proses rawatan air sisa terbaik mengikut ciri air sisa turut dibincangkan.

This course covers wastewater treatment processes based on local legal requirements, sources and characteristics of wastewater and the role of treatment plants. The basic parameters of testing the effectiveness of plant operations, process monitoring as well as plant health and safety issues are emphasized. The process best wastewater treatment based on the characteristics of wastewater is also discussed.

ESC3208 Rawatan Air Minum / Potable Water Treatment**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi isu berkaitan kepentingan sumber bekalan air, kesan pencemaran air dan kaedah rawatan air minum. Parameter penting penentuan kualiti air dan kaedah nyahjangkitan air termasuk kesan kualiti air terhadap kesihatan manusia dibincangkan. Penilaian dalam keberkesanan alat rawatan air yang sesuai turut ditekankan

This course covers issues related to the importance of water supply, the effects of water pollution and drinking water treatment methods. Important parameters for determining water quality and methods of water disinfection including the effects of water quality on human health are discussed. Evaluation of the effectiveness of appropriate water treatment tools is also emphasized.

ESC3402 Geologi Alam Sekitar / Environmental Geology**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas geologi alam sekitar, proses dalaman dan permukaan bumi. Sumber mineral dan tenaga, pelupusan sisa, jenis pencemaran serta pengukuran parameter geologi ditekankan. Isu terkini geologi alam sekitar dan langkah pencegahannya turut dibincangkan.

This course covers the basic environmental geology principles, earth internal and surface processes. Mineral and energy resources, waste disposal, types of pollution, as well as geological parameters are discussed. The current issues in environmental geology and mitigation measures are also discussed.

ESC3512 Kecerdasan Buatan (AI) untuk Kelestarian Alam Sekitar / Artificial Intelligence (AI) for Environmental Sustainability**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan kaedah aplikasi Kecerdasan Buatan (AI) dalam kelestarian alam sekitar. Kaedah manipulasi Kecerdasan Buatan (AI) dalam pengurusan alam sekitar dibincangkan. Aplikasi Kecerdasan Buatan (AI) dalam penyelesaian permasalahan alam sekitar turut ditekankan.

This course covers the principles and application techniques of Artificial Intelligence (AI) in environmental sustainability. Manipulation on Artificial Intelligence (AI) in environmental management is discussed. Application of Artificial Intelligence (AI) in solving the environmental problems are discussed.

ESC3513 Pengenalan kepada Envirometriks / Introduction to Environmetrics 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi topik asas statistik untuk menganalisa data alam sekitar. Kaedah memanipulasi data statistik alam sekitar turut diperkenalkan. Aplikasi environmetriks dalam penyelesaian masalah alam sekitar juga dibincangkan.

This course covers basic topics of statistics to analyze environmental data. Methods of manipulating environmental statistics data are also introduced. The applications of environmetric in environmental problem solving are also discussed.

ESC3514 Matematik Gunaan untuk Alam Sekitar / Applied Mathematics for the Environment 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas matematik yang digunakan dalam menganalisis masalah berkaitan alam sekitar. Aplikasi persamaan dan rumus matematik dalam penyelesaian terhadap masalah alam sekitar juga dibincangkan.

This course covers the basic principles of mathematics used in analyzing environmental problems. The applications of mathematical equations and formulas in solving environmental problems are also discussed.

ESC3515 Envirometriks / Environmetrics 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi kaedah analisis dan interpretasi data alam sekitar dalam langkah tebatan sumber pencemaran. Kajian kes digunakan untuk mengesahkan model ramalan berdasarkan analisis masa sebenar. Kaedah analisis multivariat dan pemodelan untuk menilai kualiti alam sekitar juga dibincangkan.

This course covers methods in analysis and interpretation of environmental data in mitigating pollution sources. Case studies are used to validate prediction models based on real time analysis. Multivariate analysis and modelling methods to evaluate environmental quality are also discussed.

ESC3516 Sistem Maklumat Geografi dan Penderiaan Jarak Jauh Alam Sekitar / Environmental Geographic Information System and Remote Sensing 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi pemahaman asas dalam konsep Sistem Maklumat Geografi (GIS) dan penderiaan jarak jauh alam sekitar. Asas geoinformatik dan struktur asas data geospasial turut ditekankan. Prinsip penderiaan jarak jauh dan kaedah asas pemprosesan digital penderiaan jarak jauh seperti pembetulan imej serta pengklasifikasi spektral untuk klasifikasi tematik juga dibincangkan.

This course offers insights into the basic concepts of environmental geographic information system and remote sensing. The basic of geoinformatics concept and structure of geospatial data is also discussed. Remote sensing principles and basic method of digital image processing such as image correction and spectral classification for thematic classification are discussed.

ESC3517 Pengurusan dan Sistem Pelaporan Projek Alam Sekitar / Environmental Project Management and Reporting System 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi perancangan dalam pengurusan projek alam sekitar secara sistematik iaitu pada tahap pra-projek, semasa dan selepas projek selesai. Kemahiran dalam menyediakan sistem pelaporan projek alam sekitar yang holistik serta mengikut garis panduan dan piawaian berkaitan juga ditekankan.

This course covers systematic planning in environmental project management, from pre-, during and post-project stages. The skills in holistic and effective environmental project reporting system according to the guideline and related standards are also emphasized.

ESC4104 Penilaian Risiko Kesihatan Alam Sekitar / Environmental Health Risk Assessment 3(3+0)

Prasyarat: Lulus / Pass ESC3102 atau/or setara / equivalent

Kursus ini merangkumi huraian dalam proses penilaian risiko kesihatan alam sekitar. Pengukuran potensi kesan dan impak buruk ke atas kesihatan individu dan populasi disebabkan pencemaran berkaitan pekerjaan dan alam sekitar turut ditekankan.

This course covers descriptions of the environmental health risk assessment process. Measurement of potential impacts and adverse impacts on the health of individuals and populations due to occupational and environmental pollution is also emphasized.

ESC4105 Penilaian Pendedahan Pencemaran / Pollution Exposure Assessment 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi laluan agen toksik berdasarkan anatomi dan fisiologi manusia. Pengiraan bagi pendedahan dibincangkan. Penilaian bagi mengawal potensi hazard di tempat kerja dan potensi ancaman kepada manusia dan persekitaran turut dibincangkan.

This course covers the route of toxic agents based on human anatomy and physiology. Calculations for exposure to pollutants are discussed. Assessments for controlling potential hazards in the workplace and potential threats to humans and the environment are also discussed.

ESC4106 Mikrobiologi Alam Sekitar / Environmental Microbiology 3(3+0)

Prasyarat: Tiada

Kursus ini merangkumi prinsip asas, kaedah dan teknologi yang digunakan di dalam bidang mikrobiologi alam sekitar. Teknik dan teknologi dalam mikrobiologi bagi penyelesaian permasalahan alam sekitar alternatif juga turut dibincangkan secara saintifik.

This course covers the basic principles, methods and technologies used in environmental microbiology. Techniques and technologies in microbiology as alternatives to solve environmental problems are also discussed scientifically.

ESC4206 Hidrogeokimia / Hydrogeochemistry 3(2+1)

Prasyarat: Tiada

Kursus ini menerangkan ciri-ciri kimia air bawah tanah dan air permukaan serta kaedah hidrogeologi untuk menilai sumber air tanah. Ia menumpukan kepada interaksi air, batuan dan alam sekitar, termasuk tindak balas hidrogeokimia serta pencemaran air tanah. Pelajar juga akan didedahkan kepada prinsip keseimbangan kimia, teknik analisis air, dan kemahiran makmal bagi kegunaan dalam pengurusan sumber air dan penyelidikan alam sekitar.

This course covers the chemical characteristics of groundwater and surface water, as well as hydrogeological methods for assessing groundwater resources. It focuses on the interactions between water, rocks, and the environment, including hydrogeochemical reactions and groundwater contamination. Students will also be exposed to principles of chemical equilibrium, water analysis techniques, and practical laboratory skills for application in water resource management and environmental research.

ESC4207 Pengurusan Lembangan dan Kualiti Air Bersepadu / Integrated Water Quality and Watershed Management 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi pendekatan pengurusan sumber menggunakan lembangan sebagai unit pengurusan dalam penentuan ciri semulajadi lembangan dan impak perubahan dalaman dan luaran dalam menentukan respons. Alat pengukuran dan pemodelan turut dibincangkan bagi membolehkan pengurusan secara komprehensif dan bersasar dapat digunapakai dan proses pembuatan keputusan termaklum.

This course includes a resource management approach using the basin as a management unit in determining the natural characteristics of the basin and the impact of internal and external changes in determining the response. Measurement and modeling tools were also discussed to enable comprehensive and targeted management to be used and the informed decision-making process.

ESC4208 Ekosistem Air Tawar Lentik dan Tanah Lembap / Lentic Freshwater and Wetland Ecosystems 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi perbandingan antara ciri utama dan fungsi ekosistem air tawar lentik dan tanah lembap. Pengukuran ciri dan produktiviti ekosistem lentik dan tanah lembap ditekankan. Pengurusan mampan kedua ekosistem juga dibincangkan.

This course encompasses the comparisons between the main features and functions of lentic freshwater and wetland ecosystems. Characteristics and productivity measurements of lentic and wetland ecosystems are emphasized. Sustainable management of both ecosystems is also discussed.

ESC4209 Ekosistem Marin dan Kehidupan / Marine Ecosystem and Life 3(2+1)

Prasyarat: Tiada

Kursus ini memberikan pendedahan kepada pelajar mengenai penilaian ancaman terhadap kelestarian ekosistem marin dan survival hidupan akuatik. Pelajar akan mengumpul data kesihatan ekosistem marin melalui pengukuran kualiti air serta pemerhatian flora dan fauna laut. Selain itu, kursus ini turut membimbing pelajar dalam menyusun keutamaan nilai berkaitan pemeliharaan ekosistem laut secara bertanggungjawab, selaras dengan cabaran global yang dinamik.

This course provides students with exposure to assessing threats to the sustainability of marine ecosystems and the survival of aquatic life. Students will collect data on marine ecosystem health through water quality measurements and observations of marine flora and fauna. In addition, the course guides students in prioritizing values related to the responsible conservation of marine ecosystems, in line with dynamic global challenges.

ESC4210 Kimia Marin / Marine Chemistry 3(2+1)

Prasyarat: Tiada

Kursus ini merangkumi proses kimia persekitaran marin termasuk asas biogeokimia oksigen, komposisi garam, spesis kimia terlarut, radioisotop, elemen utama dan surih, logam berat di persisiran pantai dan laut dalam. Satu kerja lapangan diadakan sepanjang kursus ini bagi mendedahkan para pelajar cara yang betul melakukan persampelan air laut, sedimen dan jumlah pepejal terampai, disamping kaedah-kaedah terkini penyediaan sampel. Kaedah analisis terkini sampel juga turut dibincang.

This course covers the chemical processes of marine environments which include basic biogeochemistry of oxygen, salt composition, dissolved chemical species, radioisotopes, trace and major elements, heavy metals in the coastal and deep-ocean. One fieldwork activity is held during this course to expose students to the correct seawater, sediments and total suspended solid sampling, include current method on samples treatment. An advance analysis on the samples is also discussed.

ESC4304 Kimia Atmosfera dan Interaksi / Atmospheric Chemistry and Interaction**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi pelbagai aspek komposisi kimia atmosfera, kepentingan serta interaksi. Tumpuan utama adalah kepada proses dan hasil interaksi antara atmosfera dengan ekosistem termasuk tanah dan lautan serta implikasi terhadap perubahan iklim.

This course covers various aspects of atmospheric chemical composition, importance, and interactions. The main focus is on the process and results of the interaction between the atmosphere and the ecosystem including land and ocean as well as the implications for climate change.

ESC4305 Pemencilan Karbon dan Perubahan Iklim / Carbon Sequestration and Climate Change**3(2+1)**

Prasyarat: Lulus/Pass ESC3029 atau/or setara / equivalent

Kursus ini merangkumi simulasi perubahan iklim dan pemencilan karbon, dalam pengurusan alam sekitar. Analisis keseimbangan karbon, karbon bajet dalam sistem aliran karbon bagi memahami konsep pemencilan karbon dalam ekosistem dan perdagangan karbon dibincangkan. Penilaian impak perubahan iklim terhadap komuniti dan tindakan penyesuaian turut ditekankan.

This course covers climate change simulation and carbon sequestration in environmental management. The analysis of carbon balance, carbon budgets in the carbon flow system to understand the cost of carbon abatement in the ecosystem and carbon trade is discussed. Evaluation of the impact of climate change on communities and adaptive actions is also emphasized.

ESC4306 Teknologi Pengawalan Pencemaran Udara / Air Pollution Control Technology**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi pelbagai aspek sumber pencemar udara, pengukuran dan teknologi kawalan pencemaran udara. Tumpuan kepada teknologi kawalan efektif merangkumi latar belakang, konsep kekunci utama, reka bentuk dan kajian kes, serta aspek persepsi masyarakat, dasar dan perundangan turut dibincangkan.

This course covers various aspects of air pollution sources, measurement, and technology of air pollution control. Focuses on effective control technologies, including backgrounds, key concepts, case design and studies, as well as aspects of public perception, policy and legislation are also discussed.

ESC4407 Valorisasi Biojisim dalam Ekonomi Kitaran / Biomass Valorisation in Circular Economy**3(3+0)**

Prasyarat: Tiada

Kursus ini memfokuskan pada prinsip dan teknologi daripada biojisim pengakuan, menekankan kepada penukaran daripada biojisim kepada bio bahan api, biokimia, dan bahan bio. Kursus ini direka untuk menyediakan pelajar dengan praktikal persefahaman terhadap aspek alam sekitar, ekonomi, dan teknologi daripada penggunaan biojisim.

This course focuses on the principles and technologies of biomass valorisation, emphasizing biomass conversion into biofuels, biochemicals, and biomaterials. The course is designed to provide students with a practical understanding of the environmental, economic, and technological aspects of biomass utilisation.

Prasyarat: Tiada

Kursus ini merangkumi struktur, fungsi, dan interaksi antara ekosistem tanah dan hutan serta peranannya dalam menyokong keseimbangan alam sekitar. Fokus utama adalah memahami komponen utama ekosistem, seperti tanah, flora, fauna, dan mikroorganisma, serta impak aktiviti manusia terhadap kestabilan ekosistem serta pendekatan pemuliharaan yang inovatif dan lestari, termasuk penggunaan teknologi seperti GIS, fitoremediasi, dan solusi berasaskan alam.

This course covers the structure, function, and interaction between soil and forest ecosystems and their role in supporting environmental balance. The focus is to understand the main components of ecosystems, such as soil, flora, fauna, and microorganisms, as well as the impact of human activities on ecosystem stability as well as innovative and sustainable conservation approaches, including the use of technologies such as GIS, phytoremediation, and nature-based solutions.

ESC4409 Kaedah Penyiasatan Subpermukaan / Subsurface Investigation Methods**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kaedah penyiasatan dan pengenalan struktur, parameter fizikal dan bahan kandungan subpermukaan bumi untuk peninjauan dan carigali. Aplikasi pelbagai kaedah penilaian akuifer didedahkan. Penilaian pelbagai kaedah geofizik dalam peninjauan carigali turut ditekankan.

This course covers methods to investigate and identify structures, physical parameters and the constituents of the earth subsurface for surveying and exploration purposes. Application of various aquifer evaluation methods are exposed. Multiple geophysics evaluations in surveying and exploration are also discussed.

ESC4410 Teknologi Rawatan Sisa Berbahaya / Hazardous Waste Treatment Technology**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi ciri fizikal, kimia dan ketoksikan sisa berbahaya yang membentuk asas untuk pengelasan hazad dan penentuan pergerakan, taburan dan juga impak ke atas kesihatan manusia dan alam sekitar. Teknologi rawatan dan pelupusan sisa berbahaya dibincangkan. Pelaksanaan pengasingan sisa berbahaya dalam kehidupan harian turut ditekankan.

This course covers the physical, chemical and toxic properties of hazardous wastes that form the basis for hazard classification and determination of their movement, distribution as well as their impacts on human health and the environment. Treatment and disposal technologies are discussed. The implementation of segregation of hazardous waste in daily life is also emphasized.

ESC4411 Pengurusan Kawalan Hakisan Tanah, Sedimen dan Air Larian Permukaan / Soil Erosion, Sediment and Stormwater Management Control**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi perancangan dan pengurusan risiko hakisan tanah dan pengurusan sedimen dalam air larian permukaan bagi mengawal impaknya kepada alam sekitar. Sistem organisasi maklumat dalam perancangan kawalan hakisan tanah juga dibincangkan. Kaedah kawalan dan amalan mitigasi terbaik dalam projek pembangunan dan perubahan guna tanah yang berisiko dalam kawasan tropika turut ditekankan.

This course covers planning and management of soil erosion risk and management of sediment-laden runoff to control its impact on the environment. The organisation system of information on erosion control planning is discussed. Best mitigation controls and practices in risky land development and land use change projects in the tropics are also emphasized.

ESC4412 Geofizik Alam Sekitar / Environmental Geophysics**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi kajian struktur dalaman bumi serta kandungannya melalui kaedah geofizik. Hubungkait antara pencemaran akibat aktiviti manusia dan semulajadi serta langkah kawalan turut dibincangkan.

This course covers the study of the internal structure of the earth and its contents through geophysical methods. The relationship between pollution caused by human and natural activities and control measures is also discussed.

ESC4519 Respons Kecemasan Alam Sekitar / Environmental Emergency Response**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi aliran dan impak bahan pencemar, serta langkah-langkah keselamatan ketika kemalangan. Respons bagi pelbagai senario kecemasan alam sekitar dibincangkan.

This course covers the flow and impact of pollutants, as well as safety measures in the event of an accident. Responses for various environmental emergency scenarios are discussed.

ESC4520 Biologi Pencemaran / Pollution Biology**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi konsep biologi pencemaran dan ekotoksikologi. Teknik dan kaedah penilaian untuk proses pemantauan dan kawalan pencemaran, serta kaedah membeza punca dan menilai kesan pencemaran air, udara dan tanah dibincangkan.

This course covers the concepts of pollution biology and ecotoxicology. The techniques and methods of assessment for the process of monitoring and control of pollution, as well as methods of determining the causes and assessing the effects of water, air and soil pollution are discussed.

ESC4521 Pemodelan Sistem Alam Sekitar / Environmental System Modelling**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi sistem analisis untuk kajian fenomena alam sekitar yang berubah mengikut masa. Pembentukan dan ujian model sistem untuk penganalisan dan pengurusan sistem alam sekitar ditekankan. Hasil model bagi penilaian bagi membantu proses pembuatan keputusan berkaitan pengurusan alam sekitar turut dibincangkan.

This course covers analysis systems for to study the environmental phenomena that change over time. Development and testing system models for environmental system analysis and management are emphasized. The results of the model for evaluation to aid decision making in environmental management are also discussed.

ESC4522 Ekologi Industri / Industrial Ecology**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep ekologi industri serta isu pembangunan mampan menggunakan kerangka ekologi industri. Pendekatan ekologi industri dan ekonomi kitaran dalam pengurusan sumber bagi mengurangkan impak alam sekitar dibincangkan.

This course covers the concepts of industrial ecology as well as issues in sustainable development using industrial ecological frameworks. The industrial ecological approach and circular economy in resource management to reduce environmental impact are discussed.

ESC4523 Teknologi Pintar untuk Alam Sekitar / *Intelligent Technology for Environment* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep dan kaedah pembinaan sistem kepintaran buatan dan peranan dalam pengurusan alam sekitar. Faedah dan kekangan kepintaran buatan berbanding sistem konvensional dalam menangani masalah alam sekitar. Hubungkait kaedah kepintaran buatan melalui pembangunan prototaip turut didedahkan kepada pelajar

This course covers concepts and methods in the development of an artificial intelligence system and its role in environmental management. Benefits and limitations of artificial intelligence versus conventional methods in solving environmental problems are emphasized. The linkages of artificial intelligence through prototype development are also exposed to the student.

ESC4524 Forensik Alam Sekitar / *Environmental Forensics* **3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi penilaian pengangkutan, persinggahan, dan kitar hayat bahan pencemar dalam persekitaran. Aspek pengukuran bahan pencemar dalam penyiasatan alam sekitar turut dibincangkan. Selain itu, satu sesi kerja lapangan akan dijalankan bagi mengaplikasikan kajian kes berkaitan forensik alam sekitar. Penekanan juga diberikan kepada aplikasi forensik alam sekitar dalam menyelesaikan isu-isu pencemaran yang kompleks.

This course covers the assessment of the transport, fate, and life cycle of pollutants in the environment. It also discusses pollutant measurement in environmental investigations. Additionally, a fieldwork session will be conducted to apply case studies related to environmental forensics. Emphasis is also given to the application of environmental forensics in solving complex pollution issues.

ESC4525 Pemuliharaan dan Pengurusan Biodiversiti / *Biodiversity Conservation and Management* **3(3+0)**

Prasyarat: Tiada

Kursus ini meliputi kepentingan biodiversiti dan pemuliharaan terhadap kestabilan ekosistem semulajadi. Pengenalpastian konflik dan ancaman, serta strategi pemuliharaan dan pembaikpulih biodiversiti turut dibincangkan

This course covers the importance of biodiversity and conservation to the stability of the natural ecosystem. Identification of conflicts and threats, as well as conservation and rehabilitation strategies for biodiversity are also discussed.

ESC4526 Ekologi Sumber Asli / *Ecology of Natural Resources* **3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi konsep ekologi berkaitan dengan sumber asli. Kepentingan jenis sumber asli kepada manusia dan alam sekitar serta kaedah pemeliharaan dan pemuliharaan sumber asli untuk pengurusan mampan dibincangkan.

This course covers ecological concepts related to natural resources. The importance of natural resources to humans and the environment as well as the preservation and conservation of natural resources for sustainable management are discussed.

ESC4527 Bioremediasi Alam Sekitar / Environmental Bioremediation**3(2+1)**

Prasyarat: Tiada

Kursus ini meliputi pelbagai aspek remediasi tanah menggunakan konsortia mikrob berpotensi untuk penguraian sebatian toksik. Proses bioremediasi, tindakbalas biokimia dan lintasan metabolik ditekankan. Kaedah rawatan in-situ dan rawatan reaktor serta aplikasi penggunaan mikrob untuk remediasi tanah dibincangkan.

This course covers various aspects of soil remediation using potential microbial consortia for the decomposition of toxic compounds. The processes of bioremediation, biochemical reactions and metabolic pathways are emphasized. In situ treatment methods and reactor treatment as well as application of microbial applications for soil remediation are discussed.

ESC4528 Sistem Pengurusan Kesihatan dan Keselamatan Industri / Industrial Health and Safety Management Systems**3(3+0)**

Prasyarat: Tiada

Kursus ini membincangkan konsep dan pendekatan dalam pengurusan sistem kesihatan dan keselamatan industri yang kondusif. Penekanan kepada perancangan, pemantauan dan tindakan untuk menangani hazard, risiko, kemalangan dan kecemasan industri dalam pelaporan maklumat turut dibincangkan. Peranan dasar, undang-undang dan piawaian dalam pengurusan kecemasan industri dan bencana alam sekitar juga ditekankan.

This course discusses the concepts and approaches in managing a conducive occupational health and safety system. Emphasis on planning, monitoring and actions to manage industrial hazards, risks, accidents and emergencies is discussed. The role of policies, laws and standards in industrial emergency and environmental disaster management are also emphasized.

ESC4529 Sistem Maklumat Geografi dan Penderiaan Jarak Jauh Alam Sekitar Lanjutan / Advanced Environmental Geographic Information System and Remote Sensing**3(2+1)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip dan kaedah lanjutan Sistem Maklumat Geografi (GIS) dan penderiaan jarak jauh (RS) alam sekitar. Kaedah manipulasi GIS dan RS lanjutan dalam pengurusan alam sekitar dibincangkan. Aplikasi GIS dan RS dalam penyelesaian permasalahan alam sekitar turut ditekankan.

This course covers the principles and advanced techniques in environmental Geography Information System (GIS) and Remote Sensing (RS). Manipulation on advanced GIS and RS in environmental management is discussed. Application of GIS and Remote Sensing in the solving the environmental problems are discussed.

ESC4530 Inovasi Alam Sekitar dan Teknologi Hijau / Environmental Innovation and Green Technology**3(1+2)**

Prasyarat: Tiada

Kursus ini meneroka penghasilan produk inovatif atau mesra alam berasaskan teknologi hijau melalui kajian kes dan analisis perbandingan silang. Ia juga merangkumi evolusi teknologi hijau serta perbincangan tentang pembangunan, pameran inovasi, dan penyelesaian alam sekitar yang mesra alam dan kos efektif.

This course explores the production of innovative or environmentally friendly products based on green technology through case studies and cross-comparative analysis. It also covers the evolution of green technologies, and includes discussions on development, innovation exhibition, and environmentally friendly and cost-effective environmental solutions.

ESC4604 Sistem Tenaga dan Alam Sekitar / Energy Systems and Environment**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi kepelbagaian sumber tenaga, penjanaan tenaga, dan impak penjanaan dan penggunaan tenaga terhadap alam sekitar dan sosio-ekonomi. Teknologi penjanaan tenaga bersih dan hijau, polisi kerajaan dan isu-isu semasa berkaitan tenaga dibincangkan. Kesedaran dalam isu dan inovasi berkaitan pengurusan tenaga dan sumber alam sekitar turut ditekankan.

This course covers the diversity of energy sources, energy generation, and the impact of energy generation and use on the environment and socio-economy. Clean and green energy generation technologies, government policies, and current issues related to energy are discussed. Awareness and knowledge of issues and innovations related to energy management and environmental resources are also emphasized

ESC4702 Interaksi Tumbuhan dan Alam Sekitar / Plant and Environment Interactions**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi prinsip interaksi antara tumbuhan dan organisma hidup dengan alam sekitarnya. Kepentingan tumbuhan terhadap alam sekitar, perubahan iklim dan pengaruh tumbuhan terhadap organisma lain serta peranannya dalam keseimbangan alam sekitar turut dibincangkan.

This course covers the interaction principle between plant and living organisms with the environment. The importance of plant towards environment, climate change and the influence of plants on other organisms and its role in environmental balance are also discussed.

ESC4804 Keterjaminan Air dan Makanan untuk Kemampanan Alam Sekitar / Water and Food Security for Environmental Sustainability**3(3+0)**

Prasyarat: Tiada

Kursus ini merangkumi hubungan antara sumber air dan keterjaminan makanan dalam konteks kemampanan alam sekitar, dengan penekanan terhadap strategi penyelesaian dan amalan pengurusan lestari yang berfokus kepada pengurusan air secara bersepadu.

This course encompasses the interconnection between water resources and food security within the context of environmental sustainability, emphasizing solution strategies and sustainable practices that focus on integrated water resources management

ESC4805 Kajian Lapangan Alam Sekitar Bersepadu / Integrated Environmental Field Study**2(0+2)**

Prasyarat: Tiada

Kursus ini merangkumi proses pengumpulan dan pengukuran data alam sekitar di lapangan serta penilaian hasil dapatan secara sistematik terhadap kesannya kepada masyarakat. Kaedah mengorganisasi data secara bersepadu yang mengambil kira norma masyarakat bagi meningkatkan kesedaran terhadap pembangunan mampan dan gaya hidup harmoni dengan alam ditekankan. Pelaporan hasil kajian lapangan bagi penyelesaian masalah alam sekitar secara strategik juga dibincangkan.

This course encompasses the process of collecting and measuring environmental data in the field, as well as the systematic evaluation of the findings in relation to their impact on society. Emphasis is placed on integrated data organization methods that consider societal norms, aiming to enhance awareness of sustainable development and a lifestyle in harmony with nature. Strategic reporting of field study findings for the resolution of environmental issues is also discussed.

ESC4806 Dokumentari Alam Sekitar / Environmental Documentaries**3(1+2)**

Prasyarat: Tiada

Kursus ini merangkumi pengenalan kepada asas penggunaan media dalam pengulasan isu-isu semasa alam sekitar. Kaedah penghasilan dokumentari alam sekitar dan hubungkait dengan persekitaran, undang-undang dan budaya dibincangkan. Pembentangan dan kritikan terhadap isu-isu alam sekitar secara bertanggungjawab turut ditekankan.

This course covers the basics of media use in reviewing current environmental issues. Methods of producing environmental documentaries and their relationship to the environment, law, and culture are discussed. The responsible presentation and critique of environmental issues are also emphasized.

ESC4807 Komunikasi Saintifik dalam Alam Sekitar / Scientific Communication in Environment **1(0+1)**

Prasyarat: Tiada

Kursus ini merangkumi perbincangan peranan pihak berkepentingan dalam penyelesaian isu alam sekitar. Perdebatan isu alam sekitar dan proses pembuatan keputusan berkaitan isu alam sekitar di peringkat komuniti, nasional dan antarabangsa melalui aktiviti permainan peranan pihak berkuasa, pemegang taruh, pihak berkepentingan dan orang awam turut ditekankan.

This course encompasses discussion on the role of stakeholders in solving environmental issues. Debates on environmental issues and decision-making processes at the local, national, and international levels through role playing activities of authorities, stakeholders and the public are also emphasized.

ESC4902 Latihan Industri / Industrial Training**12(0+12)**

Prasyarat: Telah melengkapkan dan lulus semua kursus termasuk pakej ELEX / Have completed and passed all courses including the ELEX package

Kursus ini melibatkan 24 minggu latihan industri yang memberi pendedahan langsung kepada persekitaran kerja sebenar. Pelajar melibatkan diri dalam aktiviti kajian, penyusunan dan analisis data serta pengamalan nilai profesional berdasarkan rangka kerja etika dalam konteks pekerjaan. Penekanan diberikan kepada pendedahan asas dalam teknologi, inovasi, dan peranan profesional muda dalam menyokong industri mampan selaras dengan SDG 9.5.

This course comprises 24 weeks of industrial training, providing direct exposure to real-world working environments. Students engage in research activities, data organization and analysis, as well as the application of professional values guided by an ethical framework in the workplace context. Emphasis is placed on foundational exposure to technology, innovation, and the role of young professionals in supporting sustainable industries, in line with SDG 9.5.

EST4959 Disertasi Bacelor / Bachelor Dissertation**6(0+6)**

Prasyarat: Tiada

Kursus ini merangkumi pembangunan projek penyelidikan saintifik secara berdikari dan kritis. Ia memberi tumpuan kepada penghasilan penulisan saintifik, termasuk cadangan kajian, laporan kemajuan, dan disertasi akhir. Kursus ini juga merangkumi pembentangan cadangan kajian dan penemuan hasil penyelidikan dalam meningkatkan kemahiran komunikasi dan pembelajaran sepanjang hayat.

This course covers the independent and critical development of scientific research projects. It focuses on production of scientific writing, including proposals, progress reports, and a final dissertation. The course also incorporates presentations of research proposals and research findings to enhance communication and lifelong learning skills.